

Developing Ethical Strategies for Responsible Generative AI



Course Development | Training | Agentic AI Automation

Learner Guide Slides

Version: 1.0

Course Code: TGS-2026061329

Website: www.tertiarycourses.com.sg

Digital Attendance (Mandatory)

- It is mandatory for you to take both AM, PM and Assessment digital attendance for WSQ funded courses.
- The trainer or administrator will show you the digital attendance QR code generated from SSG portal.
- Please scan the QR code from your mobile phone camera and submit your attendance.

About the Trainer

Let's Know Each Other...



Ground Rules

- Set your mobile phone to silent mode
- Actively participate in the class. No question is stupid.
- Respect each other views. Agree to disagree.
- Each person should only speak one at a time.
- Be punctual. Back from breaks on time.
- Exit the class silently if you need to step out for phone call, toilet break etc.
- 75% attendance is required for WSQ funding eligibility.

Skills Framework

- TSC Title: Responsible AI and Generative AI Practices
- TSC Code: ICT-INT-0055-1.1
- TSC Description: Responsible AI and Generative AI Practices
- Proficiency Level: Level 1
- Learning Outcomes:
 - LU1 (Generative AI Ethics Analysis)
 - LO1: This learning outcome enables participants to critically examine and analyze the societal impacts and ethical risks a...
 - LU2 (Generative AI Ethics Evaluation)
 - LO2: This learning outcome focuses on developing participants' ability to evaluate complex ethical trade-offs inherent in ...

Knowledge & Ability Statements

● LU1: Generative AI Ethics Analysis

- K2: Implications of biased AI algorithms and potential societal impact
- A4: Analyse the ethical implications of AI algorithms (e.g., assess bias and fairness concerns)
- A8: Analyse the impact of AI on different societal groups and cultures (e.g., minority groups)

● LU2: Generative AI Ethics Evaluation

- K1: Environmental impact of generative AI use (e.g., energy consumption)
- A6: Evaluate ethical frameworks and guidelines in AI development
- A2: Evaluate the trade-offs between privacy and model performance
- A9: Assess environmental impact (e.g., estimate energy footprint)

● LU3: Generative AI Strategy Development

- K3: Ethical frameworks and guidelines for generative AI-powered tools (including legal requirements for data

Course Outline

● LU1: Generative AI Ethics Analysis

- T1: Implications of biased AI algorithms and potential societal impact
- T2: Analyse the ethical implications of AI algorithms (e.g., assess bias and fairness concerns)
- T3: Analyse the impact of AI on different societal groups and cultures (e.g., minority groups)

● LU2: Generative AI Ethics Evaluation

- T1: Environmental impact of generative AI use (e.g., energy consumption)
- T2: Evaluate ethical frameworks and guidelines in AI development
- T3: Evaluate the trade-offs between privacy and model performance
- T4: Assess environmental impact (e.g., estimate energy footprint)

● LU3: Generative AI Strategy Development

- T1: Ethical frameworks and guidelines for generative AI-powered tools (including legal requirements for data

Assessment Methods & Briefing

- Assessment: Written Assessment
- Assessment format: Open Book
- Open book assessment allows you to reference your learning materials.
- Duration: As per course schedule
- Grading: Competent / Not Yet Competent
- Assessment Rules:
 - Place phones & other materials under the table
 - No photos or recording of assessment scripts

Criteria for Funding

- Minimum attendance rate of 75% based on SSG Digital Attendance record.
- Complete the assessment and be assessed as 'Competent'.
- Complete the TRAQOM survey.
- For more information on WSQ funding:
- Visit SkillsFuture portal: www.skillsfuture.gov.sg
- Eligible individuals may use SkillsFuture Credit to offset course fees.

LMS

You can access more resources from our LMS
<https://ai-lms-tms.tertiaryinfo.tech/>

Login using email and OTP or password.

LO1 | LU1 | T1: Implications of biased AI algorithms and potential societal impact

What is Implications of biased AI algorithms and potential societal impact?



Healthcare Bias

Favors certain demographic groups



Employment Bias

Discriminates in hiring processes



Criminal Justice

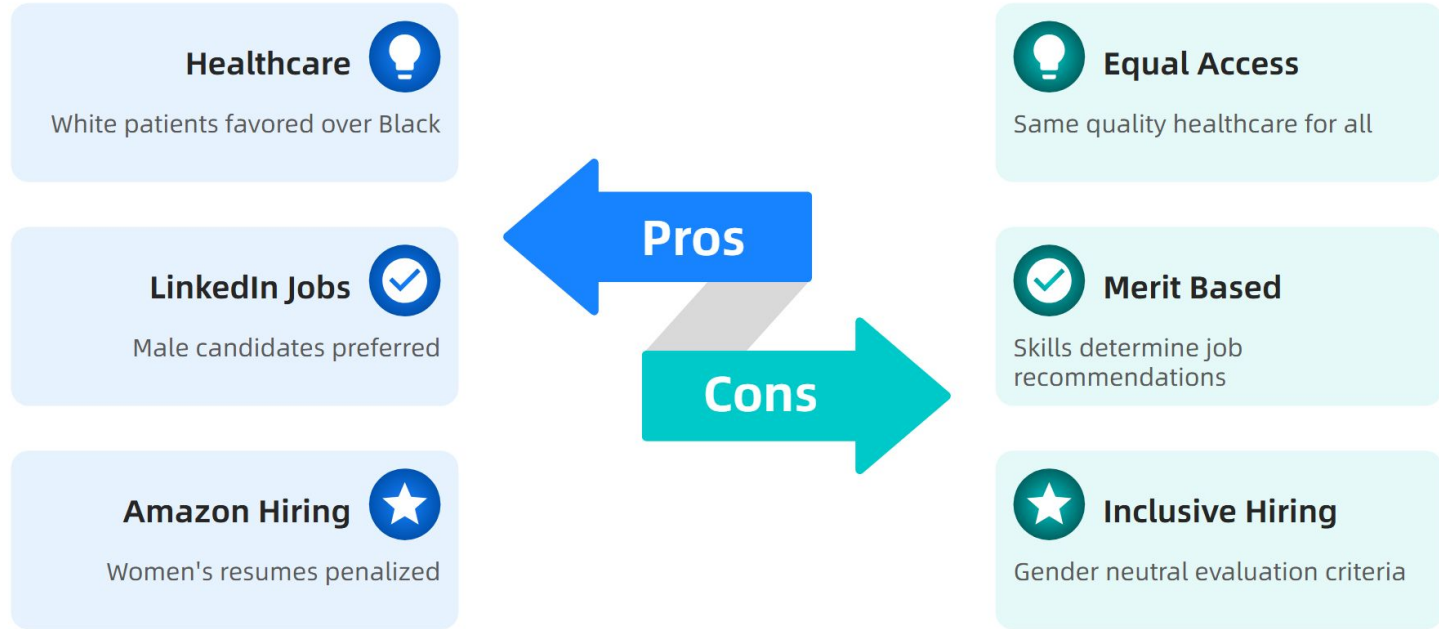
Creates unfair conviction patterns



Financial Bias

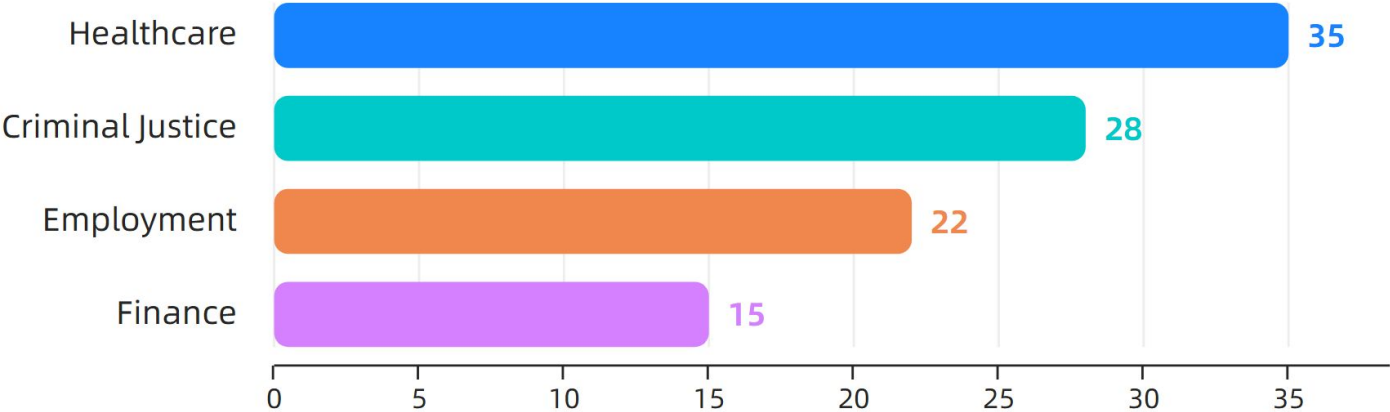
Enables discriminatory lending practices

Real-World Examples of AI Discrimination



Source: Brookings, 2023

AI Bias Impact by Sector



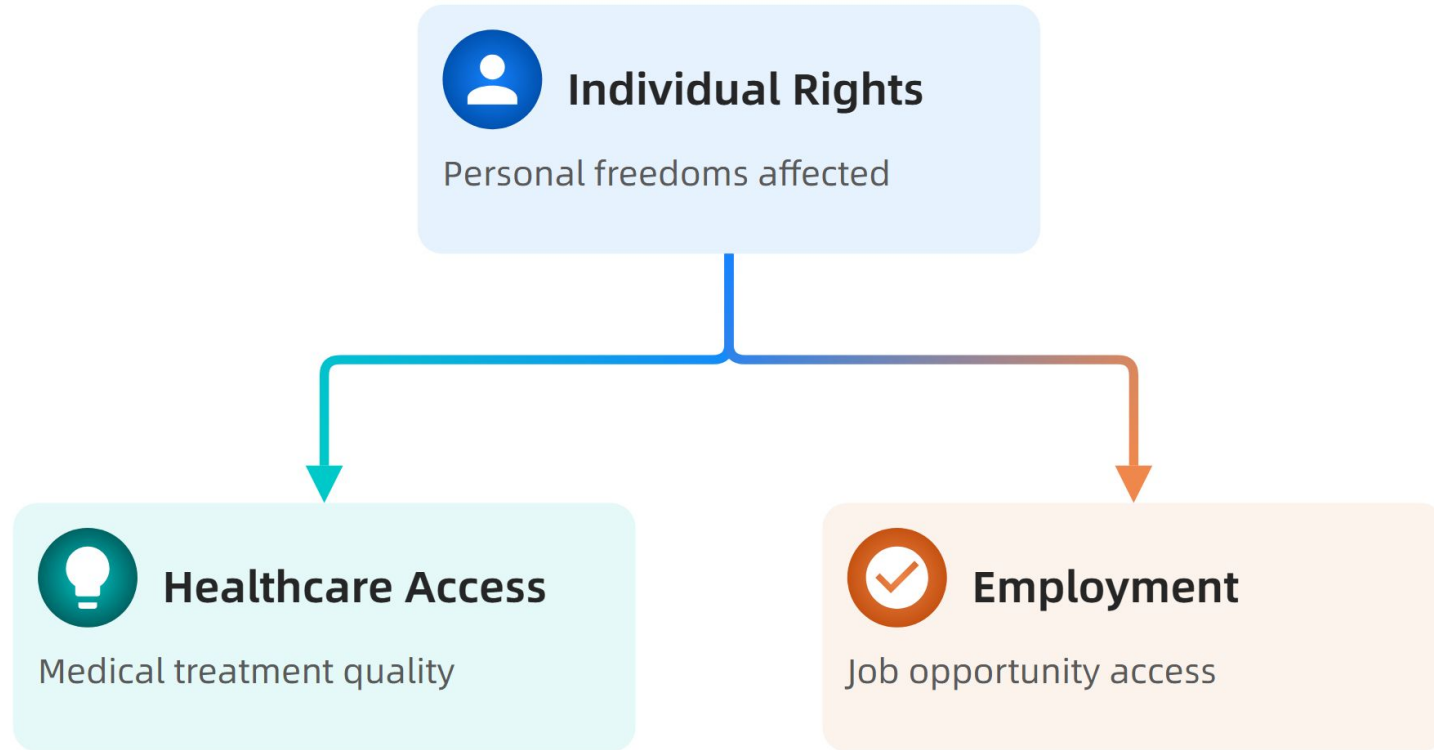
Source: MDPI, 2024

Bias Mitigation Process

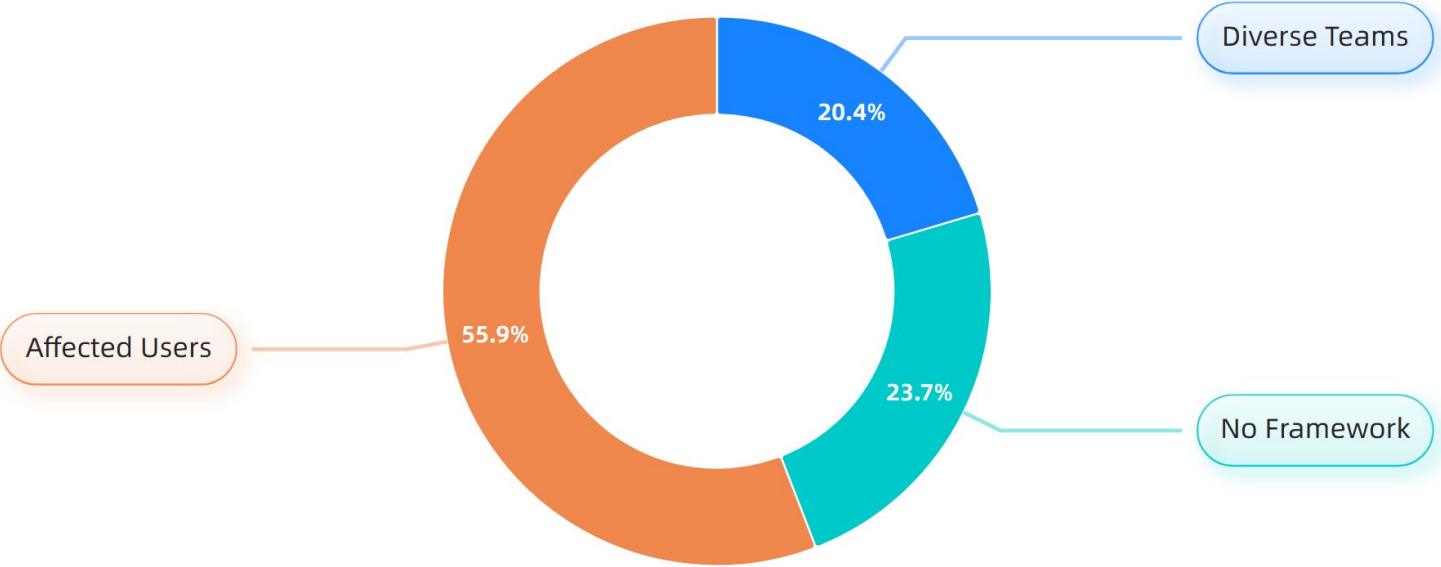


Source: NIST, 2022

Societal Impact Categories



Mitigation Success Statistics



Source: WEF, 2021; NIST, 2022

Key Takeaways for Responsible AI



Data Diversity



Team Diversity



Continuous Monitoring



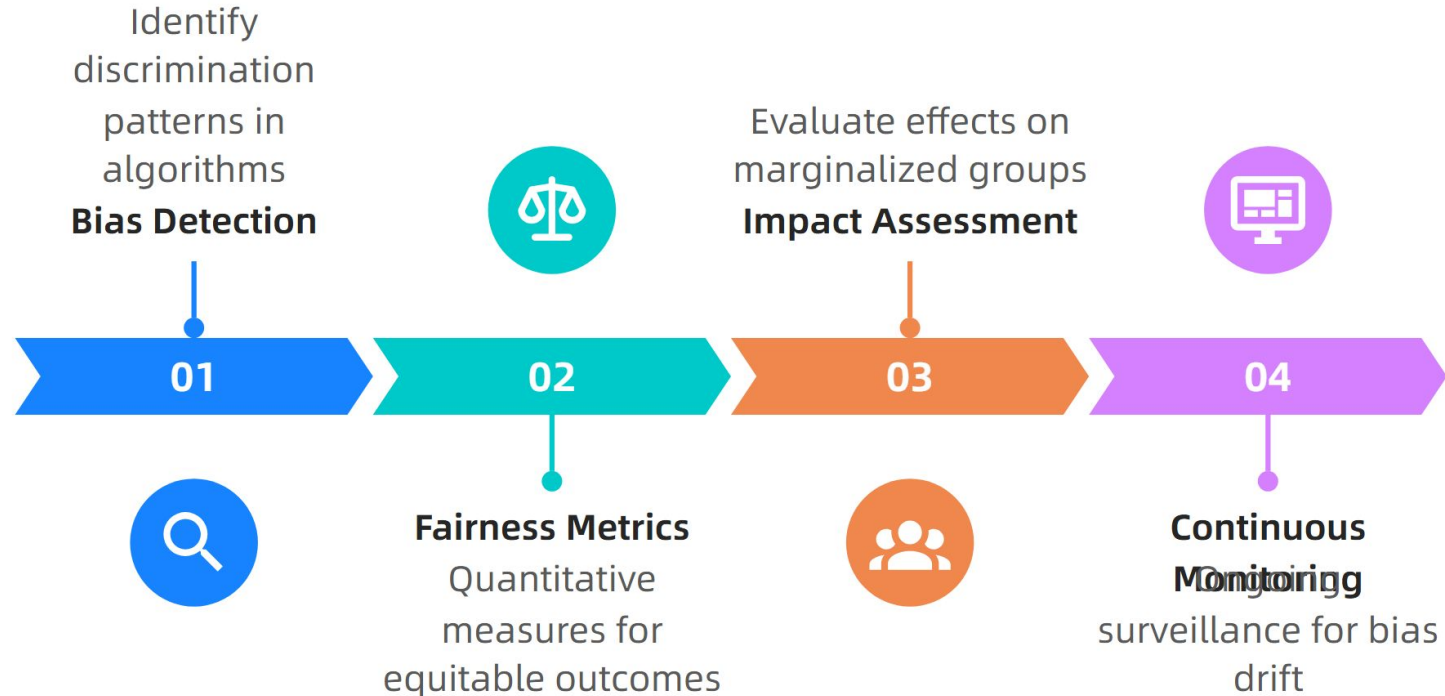
Governance Framework

Activity: Implications of biased AI algorithms and potential societal impact

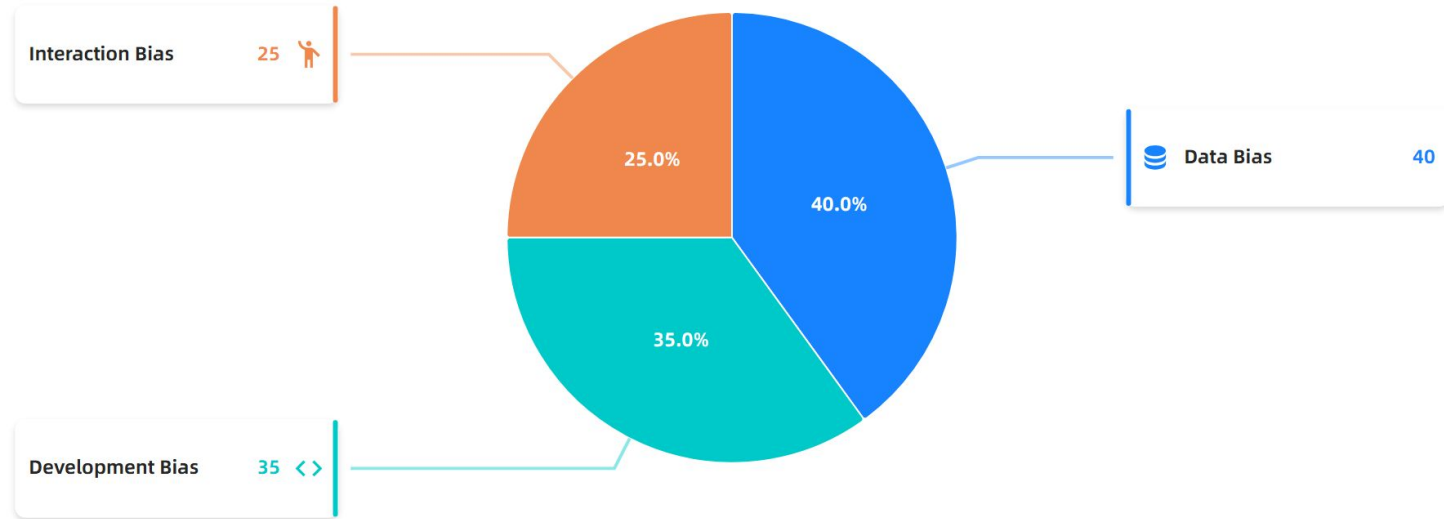
- Activity: AI Bias Impact Assessment
- Scenario: Your organization is implementing a new AI recruitment system. Recent reports suggest potential gender bias in candidate screening. You must evaluate the system's fairness and recommend mitigation strategies.
- Step 1: Analyze the recruitment AI's training data for gender representation
- Step 2: Test the system with identical resumes differing only by gender indicators
- Step 3: Document bias patterns and measure discrimination severity
- Step 4: Propose specific mitigation strategies with implementation timelines
- Expected Output: Bias assessment report with quantified discrimination metrics and actionable mitigation plan
- Duration: 25 minutes

LO1 | LU1 | T2: Analyse the ethical implications of AI algorithms (e.g., assess bias and fairness concerns)

What is Analyse the ethical implications of AI algorithms (e.g., assess bias and fairness concerns)?

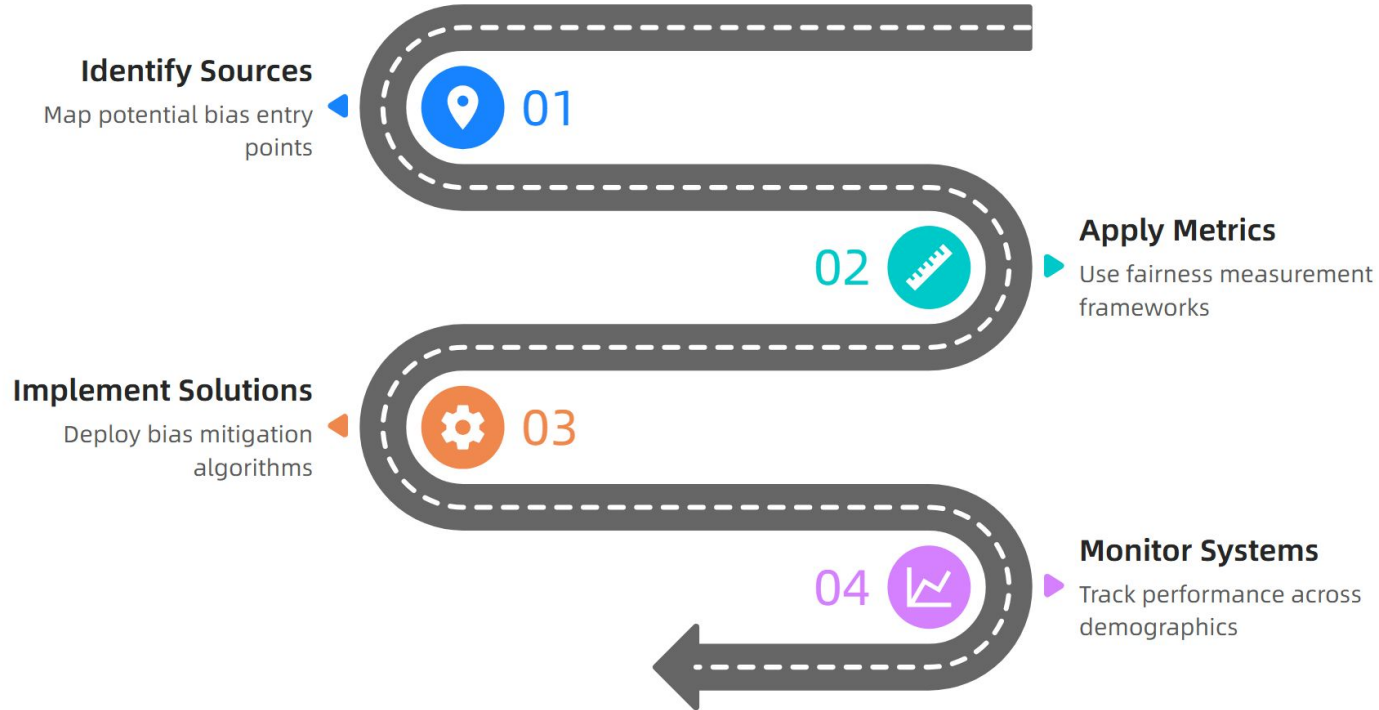


Types of AI Bias Sources



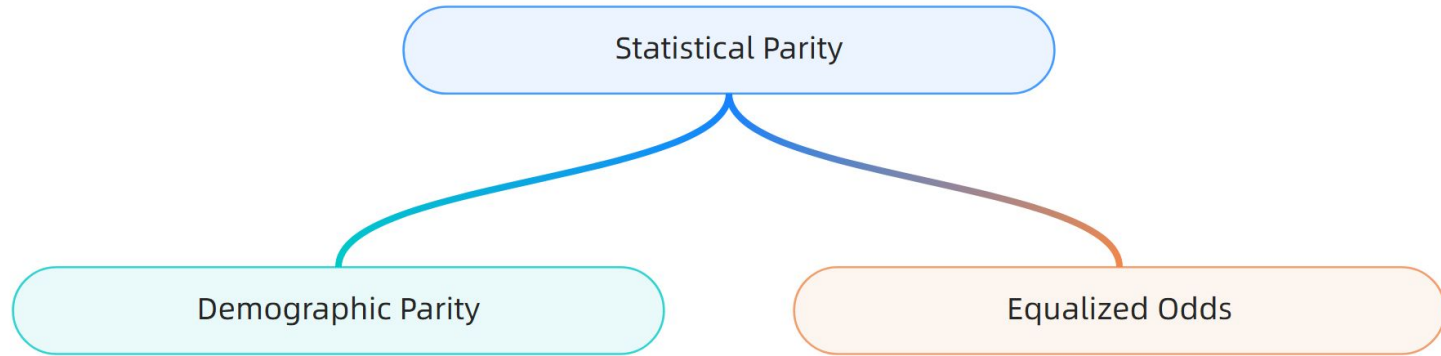
Source: Academic Research on AI Fairness, 2024

Bias Detection and Mitigation Process

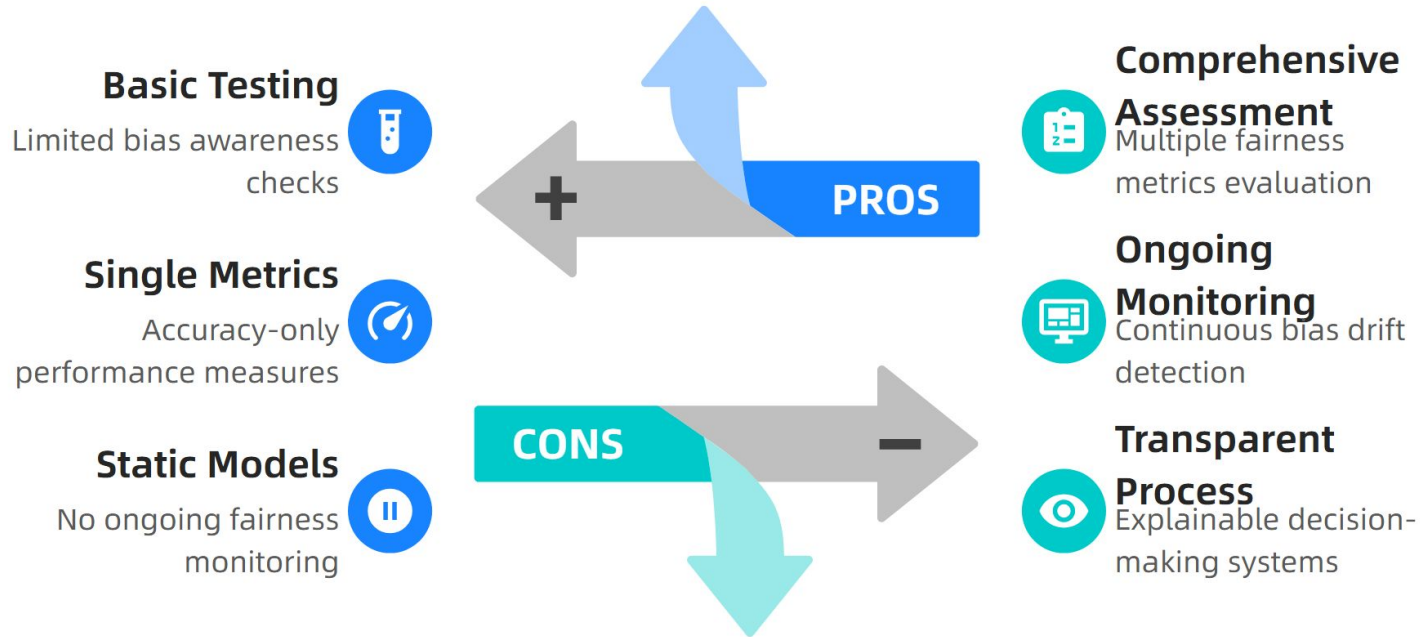


Source: IBM AI Fairness 360 Methodology, 2024

Key Fairness Metrics for Assessment

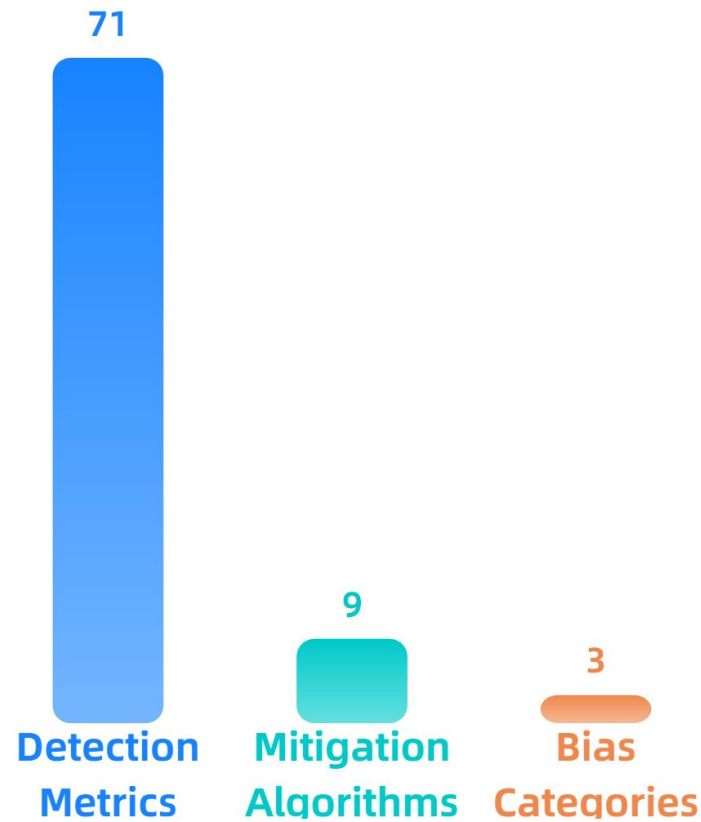


Traditional vs Ethical AI Approaches



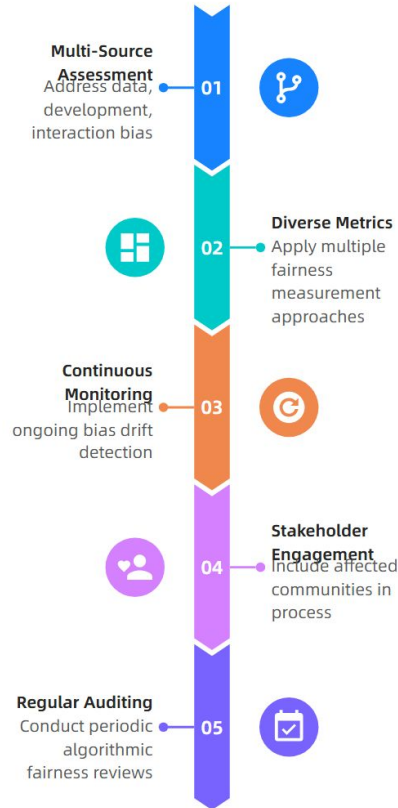
Source: Unified AI Bias Management Framework, 2023

AI Fairness 360 Toolkit Capabilities



Source: IBM AI Fairness 360 Documentation, 2024

Key Takeaways for Ethical AI Implementation



Source: UNESCO and IBM Best Practices, 2024

Activity: Analyse the ethical implications of AI algorithms (e.g., assess bias and fairness concerns)

- Activity: Bias Assessment Case Study
- Scenario: Your organization is implementing an AI-powered recruitment system. Initial testing reveals potential gender bias in candidate screening, with female applicants receiving 15% lower scores for technical roles despite equivalent qualifications.
- Step 1: Identify the three types of bias present in this scenario
- Step 2: Select appropriate fairness metrics to quantify the discrimination
- Step 3: Design a mitigation strategy using available tools and frameworks
- Step 4: Develop a monitoring plan to prevent future bias drift
- Expected Output: A comprehensive bias mitigation plan including specific metrics, tools, and ongoing monitoring procedures
- Duration: 25 minutes

**LO1 | LU1 | T3: Analyse the impact of AI
on different societal groups and cultures
(e.g., minority groups)**

What is AI Impact on Minority Communities?

Healthcare Gaps



Misdiagnosis in darker skin tones

Finance Barriers



Loan overcharging for minorities

Employment Bias



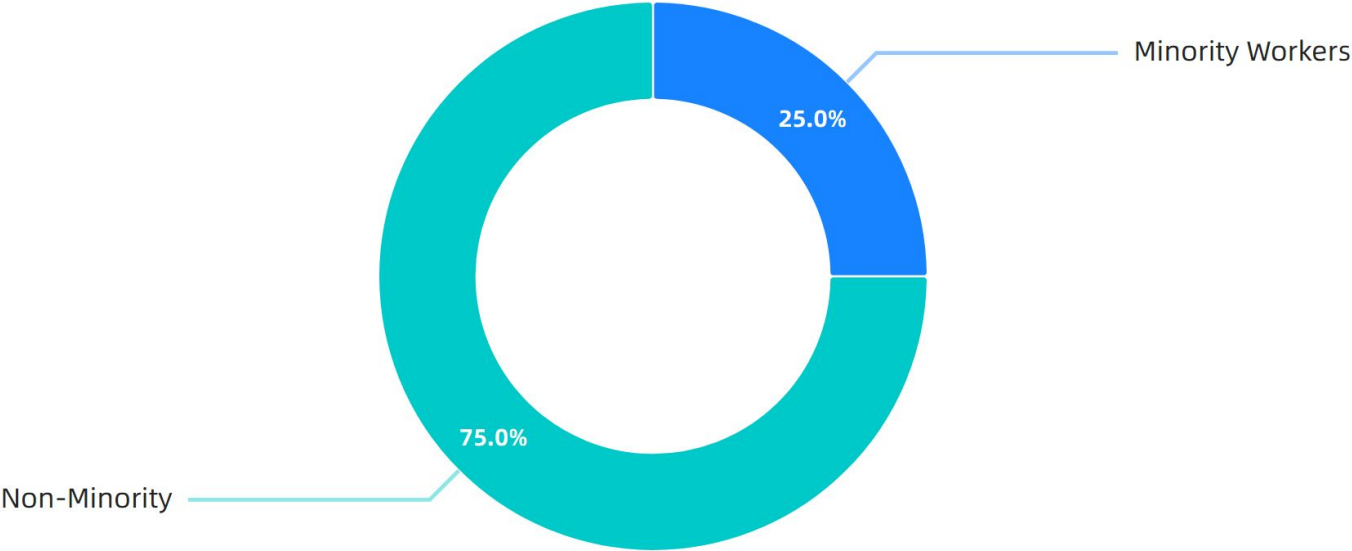
Systematic downgrading of applications

Justice Issues



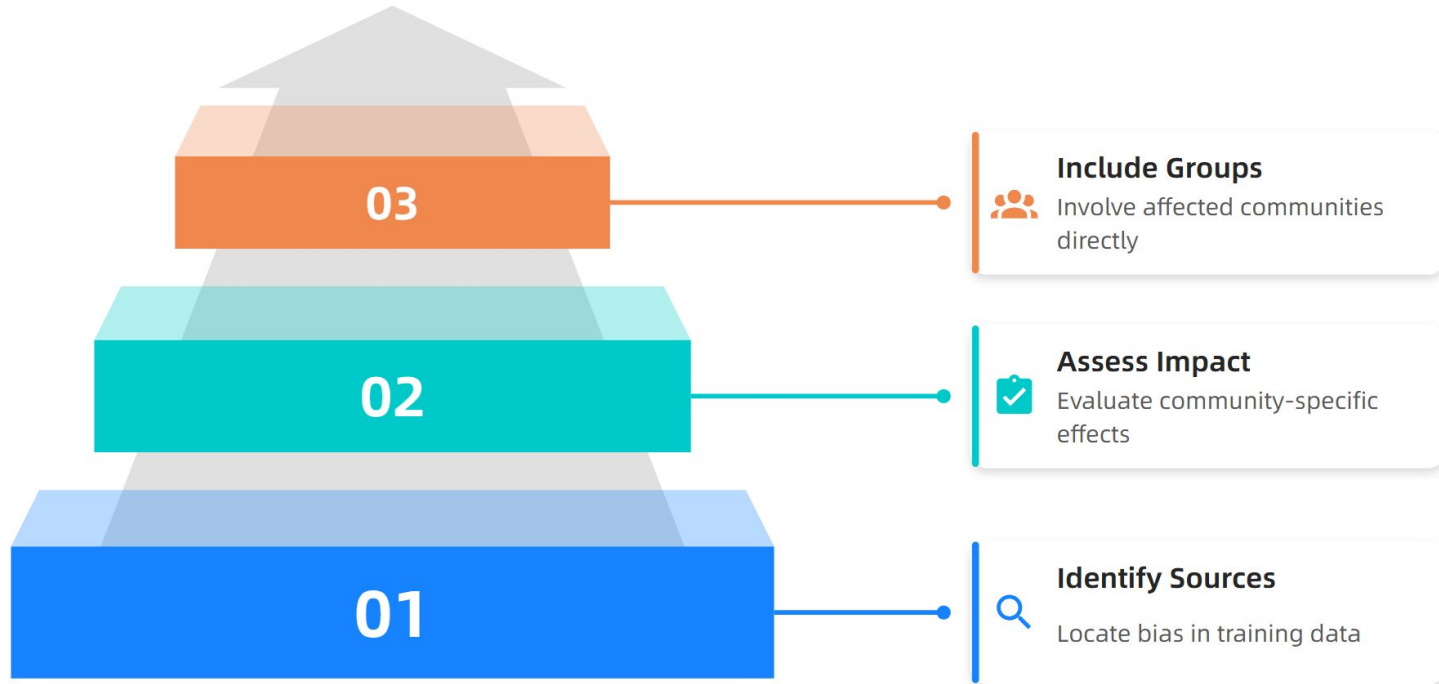
Over-policing feedback loops created

Current AI Development Demographics



Source: PMC, 2024

Inclusive AI Development Process



Source: Inclusive AI Practices, 2024

Traditional vs Inclusive AI Approaches

Homogeneous Teams

Limited diversity in development

Biased Datasets

Historical discrimination embedded

Limited Oversight

Minimal community involvement



Diverse Teams

Multi-cultural development groups

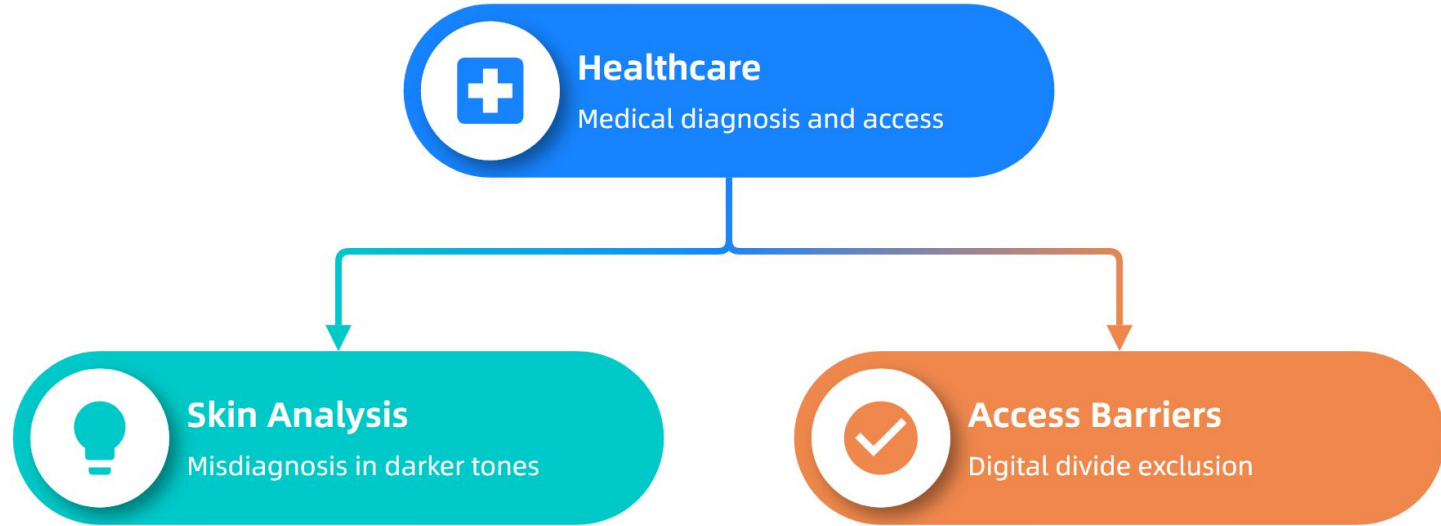
Representative Data

Balanced training datasets

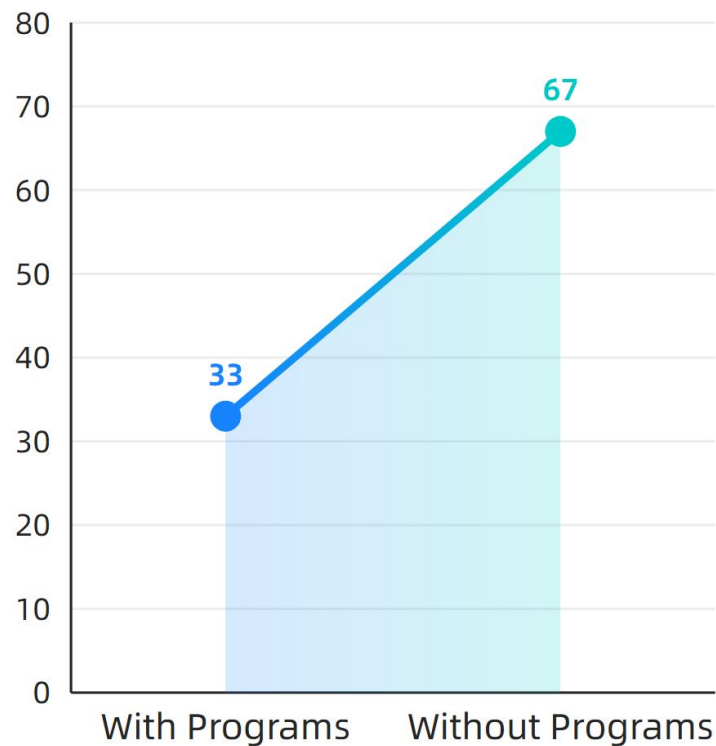
Community Input

Stakeholder engagement throughout

Sector-Specific Impact Areas



Diversity Program Implementation



Source: Industry Report, 2024

Key Takeaways for Ethical AI

Diversify Teams

Increase minority
representation
significantly

Audit Datasets

Remove historical bias
systematically

Test Rigorously

Validate across all
demographic groups

**Monitor
Continuously**
Track real-world impact
regularly

Activity: Analyse the impact of AI on different societal groups and cultures (e.g., minority groups)

- Activity: AI Bias Impact Assessment
- Scenario: Your company is developing an AI hiring tool. Recent testing revealed potential bias against minority candidates in resume screening algorithms.
- Step 1: Analyze the training dataset for historical hiring bias patterns
- Step 2: Identify which demographic groups show lower success rates
- Step 3: Engage with affected communities to understand impact
- Step 4: Propose specific technical and procedural solutions
- Step 5: Design monitoring system for ongoing bias detection
- Expected Output: Detailed bias assessment report with actionable recommendations for inclusive AI development

LO2 | LU2 | T1: Environmental impact of generative AI use (e.g., energy consumption)

What is Environmental impact of generative AI use (e.g., energy consumption)?

**Energy Surge**

Eight times higher consumption

**Carbon Growth**

Massive data center emissions

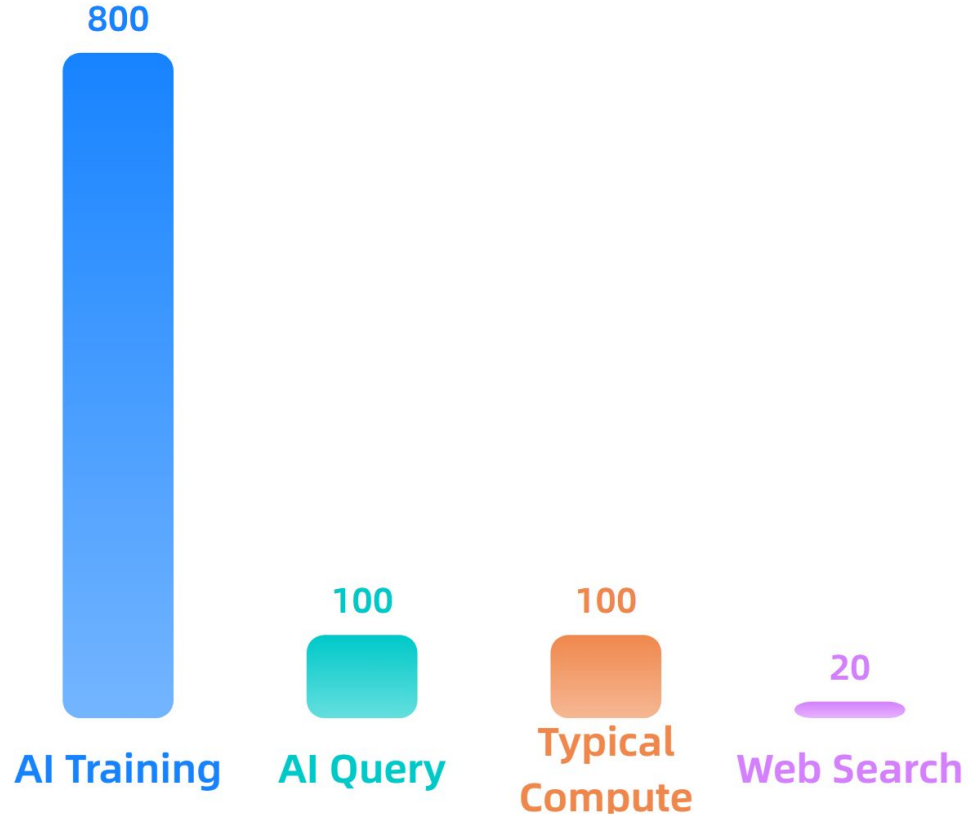
**Water Usage**

Cooling systems drain resources

**Scale Challenge**

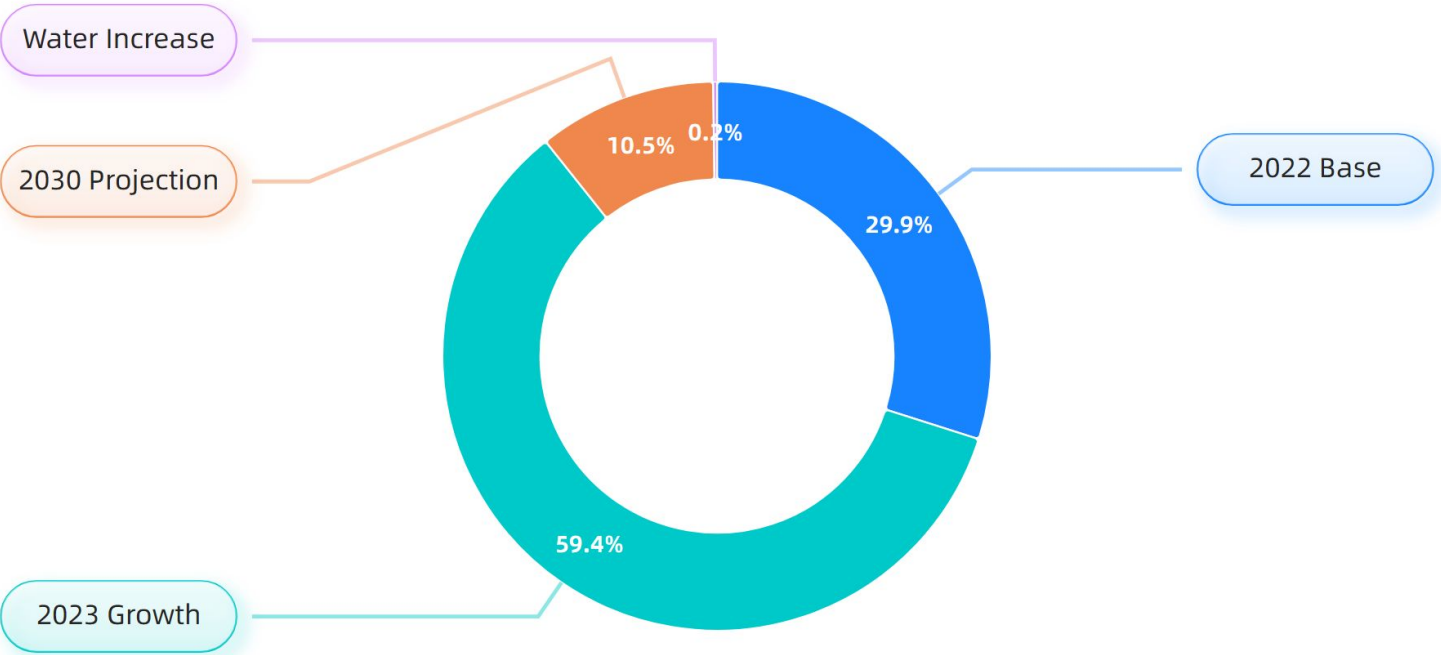
Exponential demand growth worldwide

Energy Consumption Comparison



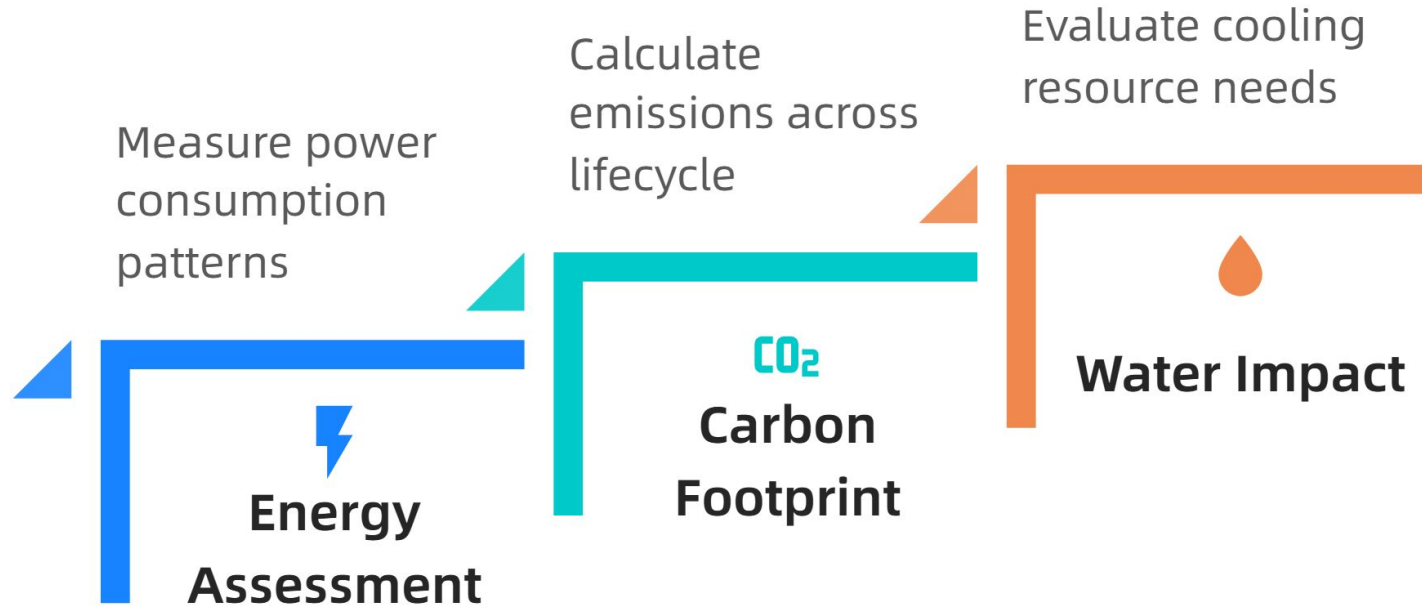
Source: MIT News, 2025

Data Center Growth Impact



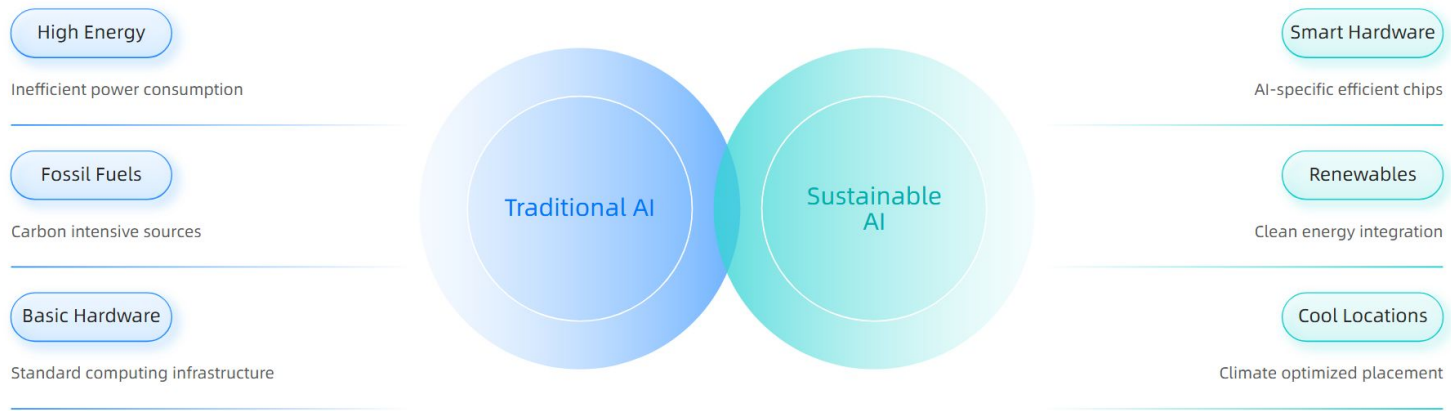
Source: Sustainable Agency, IEA, 2024

Environmental Assessment Process



Source: U.S. GAO, 2025

Sustainable vs Traditional AI



Source: MIT News, 2025

Mitigation Strategy Framework

Infrastructure

Cool Climates

AI Chips

Energy Sources

Solar Power

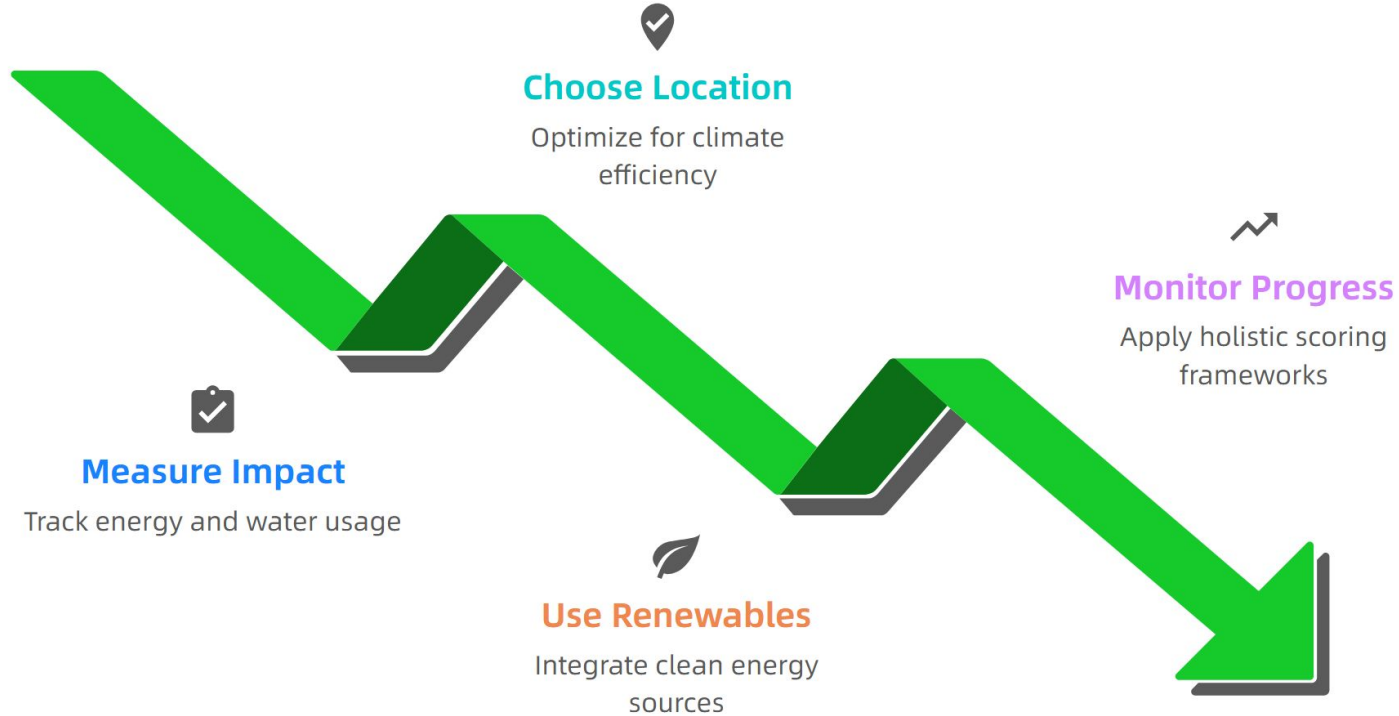
Wind Energy

Assessment

Net Score

Monitoring

Key Takeaways



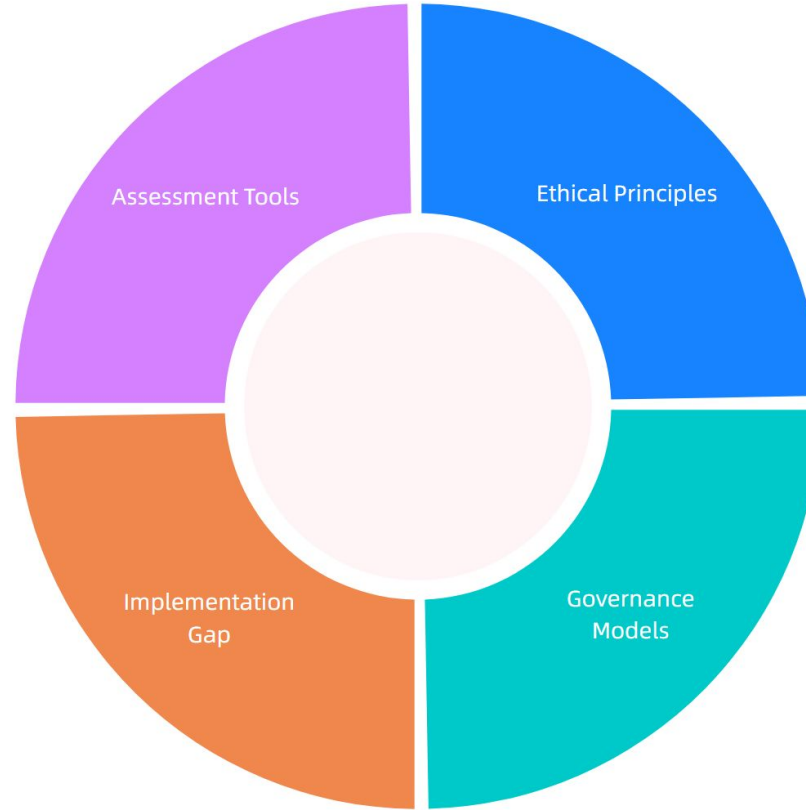
Source: Multiple sources, 2024-2025

Activity: Environmental impact of generative AI use (e.g., energy consumption)

- Activity: AI Carbon Footprint Assessment
- Scenario: Your organization is deploying a generative AI customer service chatbot. Calculate and compare the environmental impact of three deployment options: on-premises servers, cloud with standard energy, and cloud with renewable energy commitment.
- Step 1: Calculate baseline energy consumption for 10,000 daily queries
- Step 2: Estimate carbon emissions for each deployment option
- Step 3: Factor in water usage for cooling requirements
- Step 4: Apply Net Climate Impact Score methodology
- Step 5: Present recommendations with cost-benefit analysis
- Expected Output: Environmental impact comparison report with recommended deployment strategy

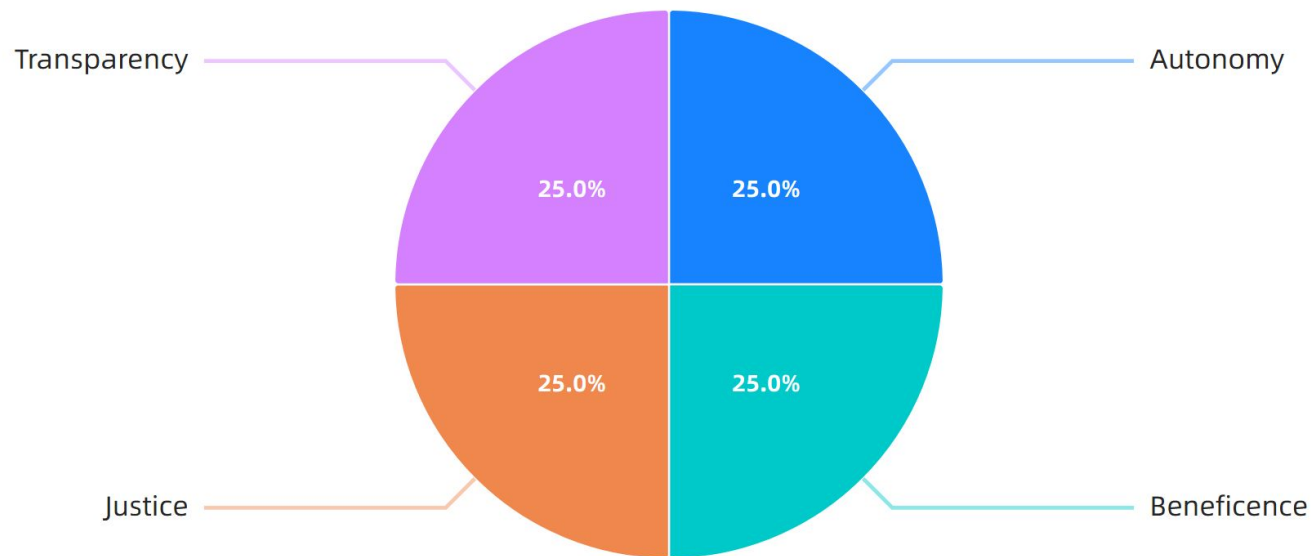
LO2 | LU2 | T2: Evaluate ethical frameworks and guidelines in AI development

What is Evaluate ethical frameworks and guidelines in AI development?

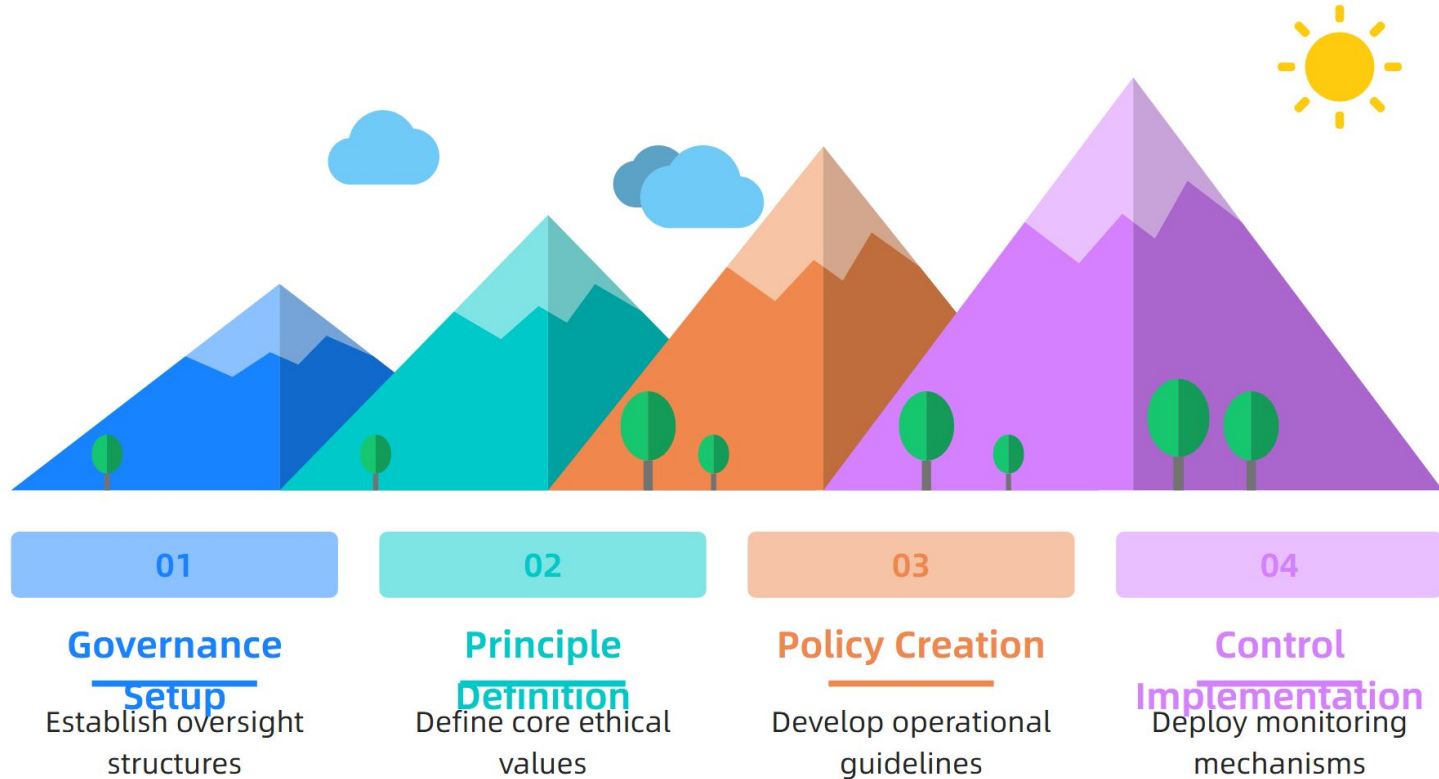


Source: UNESCO AI Ethics, 2024

Global Convergence on Core Principles



Framework Implementation Process



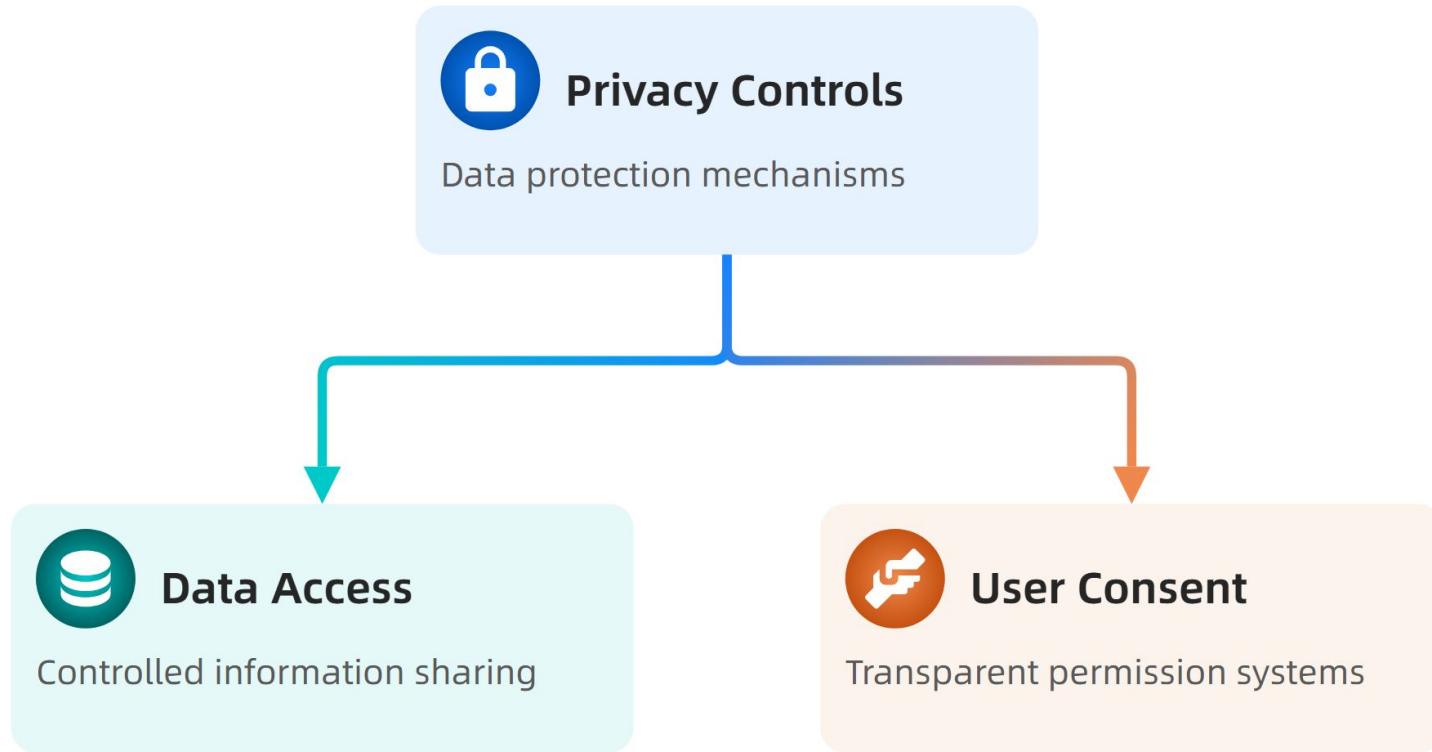
Source: Microsoft Responsible AI Standard, 2024

Leading Framework Comparison

P PRINCIPLE-BASED	P PRACTICE-BASED
● UNESCO Model ● Google Principles ● Value Foundation	● Microsoft Standard ● NIST Framework ● Metric Systems

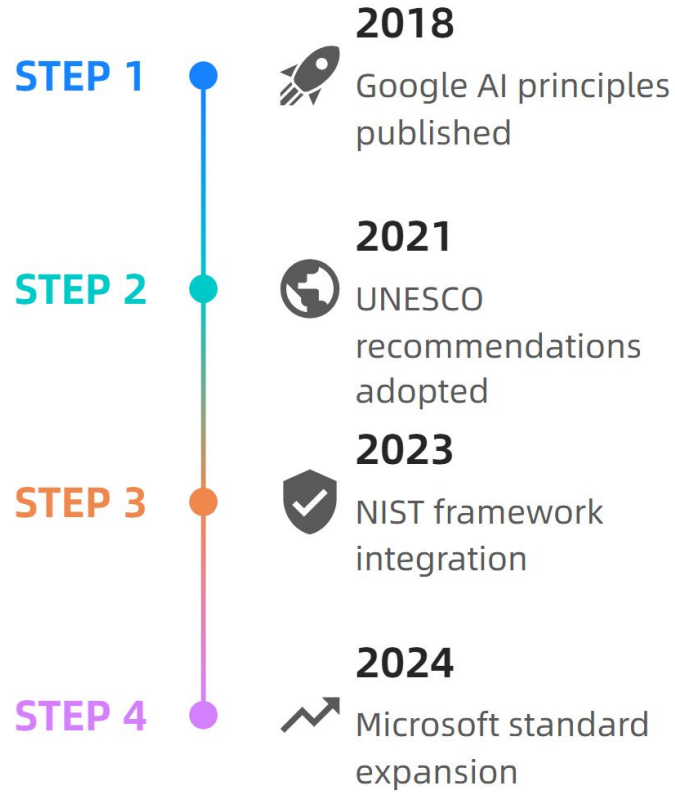
Source: Multiple framework analysis, 2024

Implementation Challenge Areas



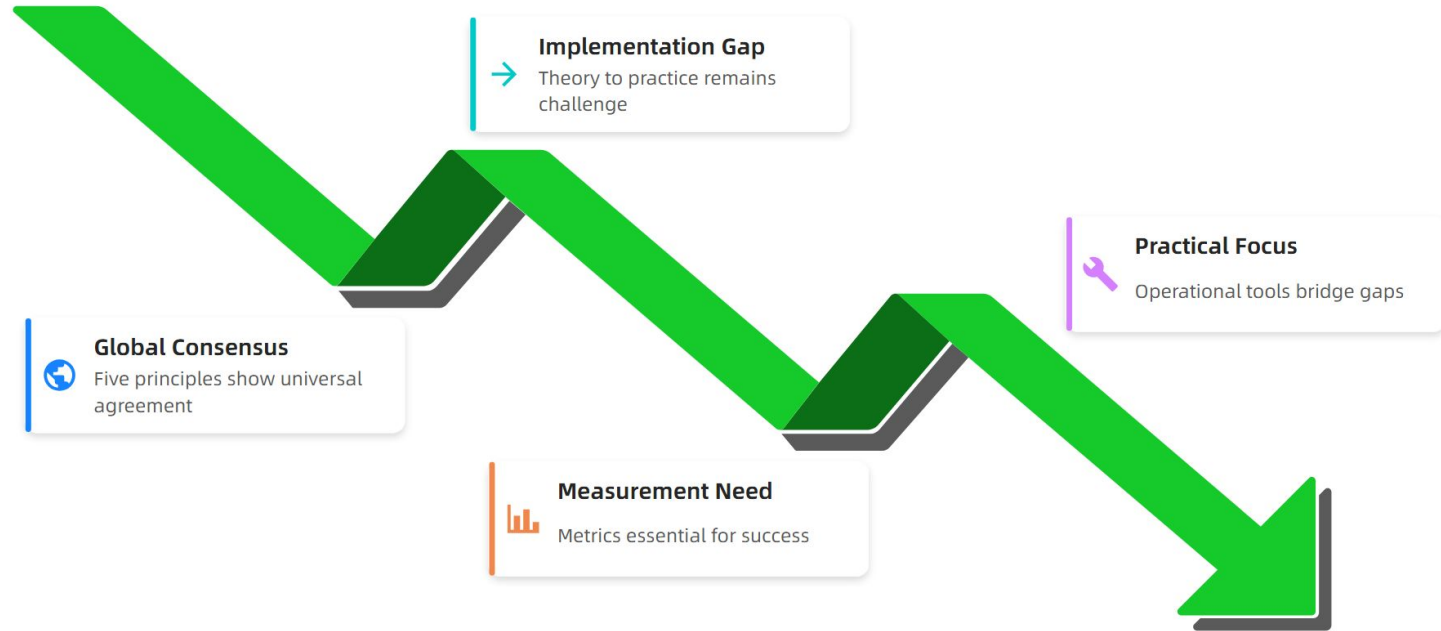
Source: Organizational challenge analysis, 2024

Framework Evolution Timeline



Source: Timeline analysis, 2018-2024

Key Framework Evaluation Takeaways



Source: Comprehensive framework analysis, 2024

Activity: Evaluate ethical frameworks and guidelines in AI development

- Activity: Framework Evaluation Exercise
- Scenario: Your organization is selecting an AI ethics framework for a new chatbot deployment. Compare UNESCO's human-rights approach with Microsoft's practice-based standard.
- Step 1: Review both framework summaries
- Step 2: Identify three key differences
- Step 3: Assess implementation feasibility
- Step 4: Recommend best approach with rationale
- Expected Output: Framework comparison matrix with implementation recommendation
- Duration: 20 minutes

**LO2 | LU2 | T3: Evaluate the trade-offs
between privacy and model performance**

What is Evaluate the trade-offs between privacy and model performance?



Privacy Risk

Data re-identification threats



Model Utility

Predictive accuracy performance



Trade-off

Inherent tension between goals



Optimization

Strategic parameter tuning

Privacy-Performance Evaluation Process

Assess Sensitivity

Evaluate data privacy requirements



Select Mechanism

Choose privacy preservation technique

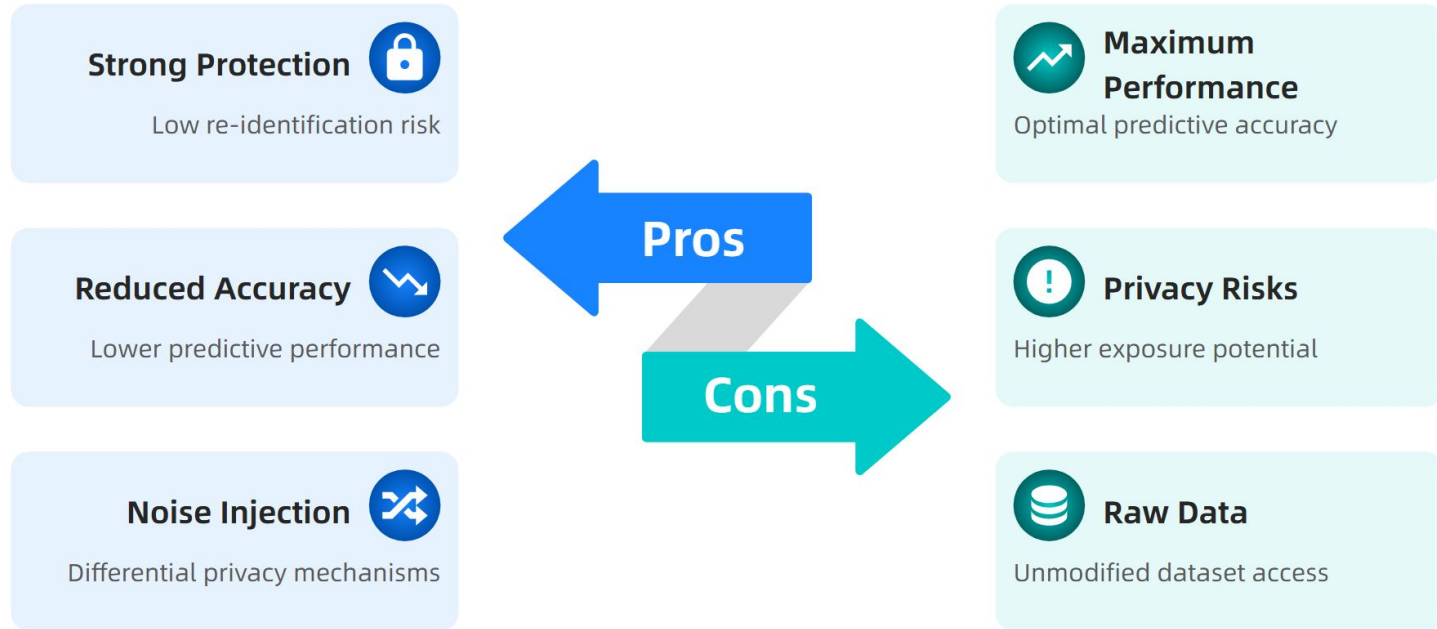


Define

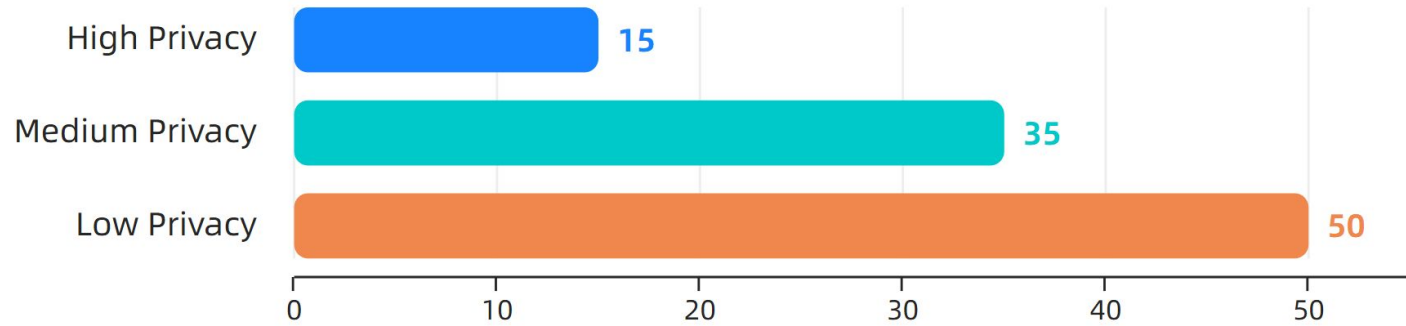
Set privacy protection standards

Requirements

Privacy vs Utility Comparison

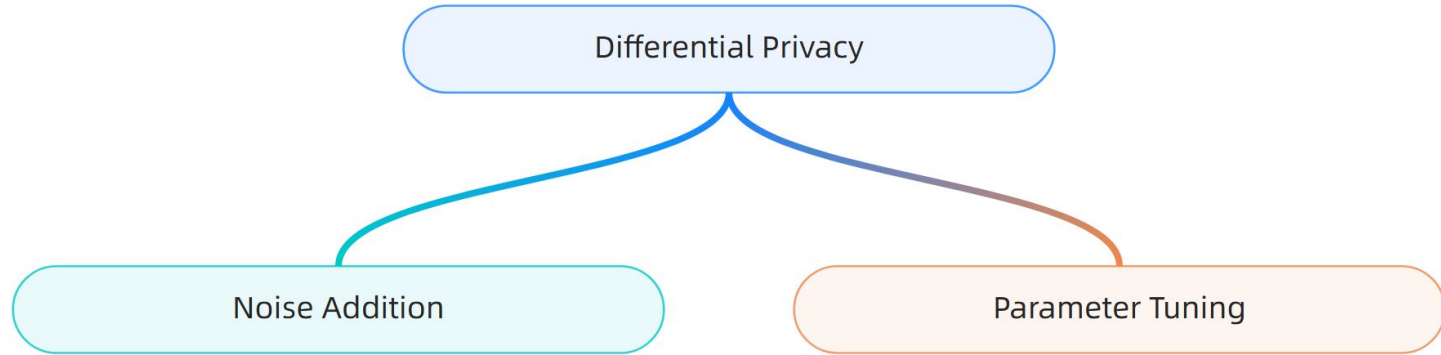


Performance Impact Statistics

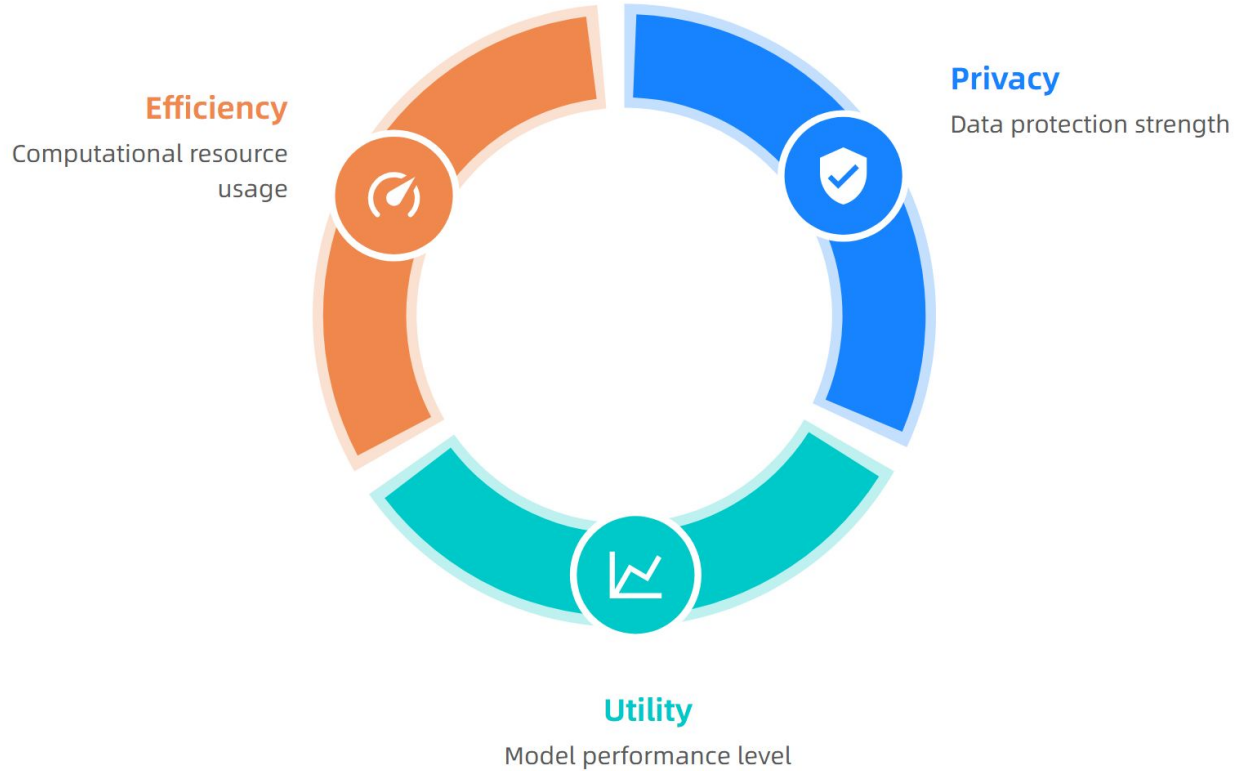


Source: ScienceDirect, 2024

Privacy Preservation Techniques

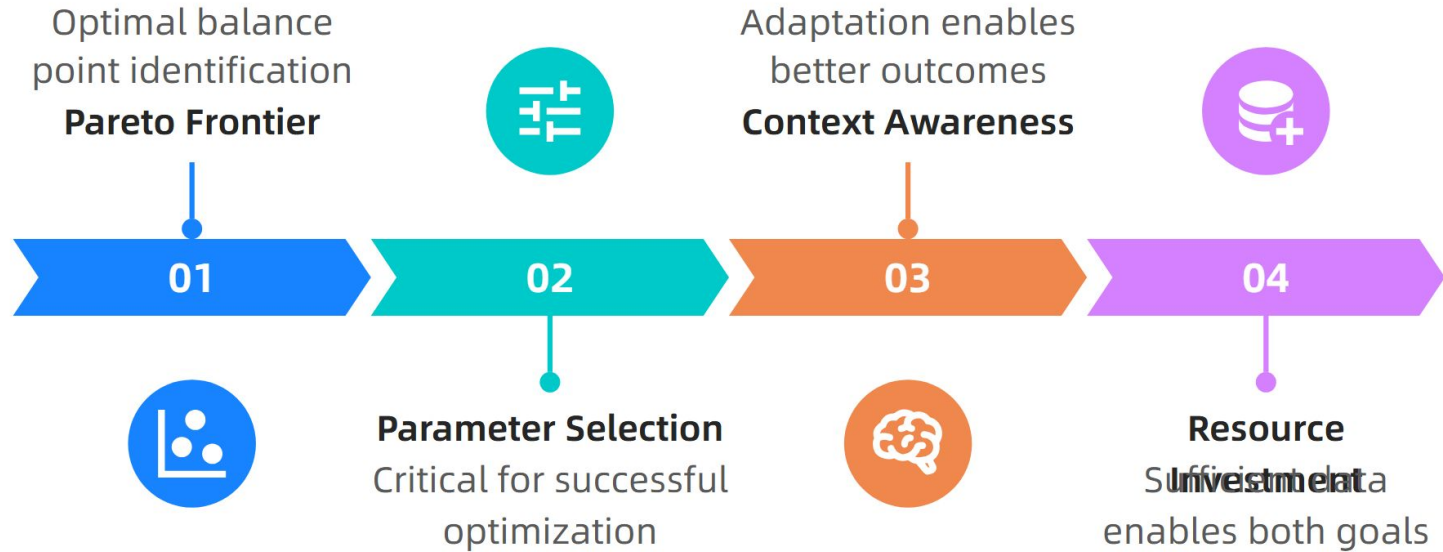


No-Free-Lunch Theorem



Source: *Trading Off Privacy Study*, 2024

Key Takeaways



Source: Research synthesis, 2024-2025

Activity: Evaluate the trade-offs between privacy and model performance

- Activity: Privacy-Performance Trade-off Analysis
- Scenario: Your organization is deploying a customer recommendation system using sensitive purchase data. The marketing team wants maximum accuracy for personalization, while the legal team requires strong privacy protection for GDPR compliance.
- Step 1: Assess the sensitivity level of customer purchase data
- Step 2: Define minimum acceptable privacy protection standards
- Step 3: Evaluate three privacy mechanisms and their utility impact
- Step 4: Select optimal parameters for chosen mechanism
- Step 5: Calculate the final privacy-utility trade-off metrics
- Expected Output: A recommendation report with privacy mechanism selection, parameter settings, and quantified trade-off analysis

LO2 | LU2 | T4: Assess environmental impact (e.g., estimate energy footprint)

What is AI Environmental Impact Assessment?

Lifecycle Analysis

Cradle to grave impact measurement



Carbon Footprint

CO₂ emissions from AI operations



Energy Tracking

Power consumption monitoring

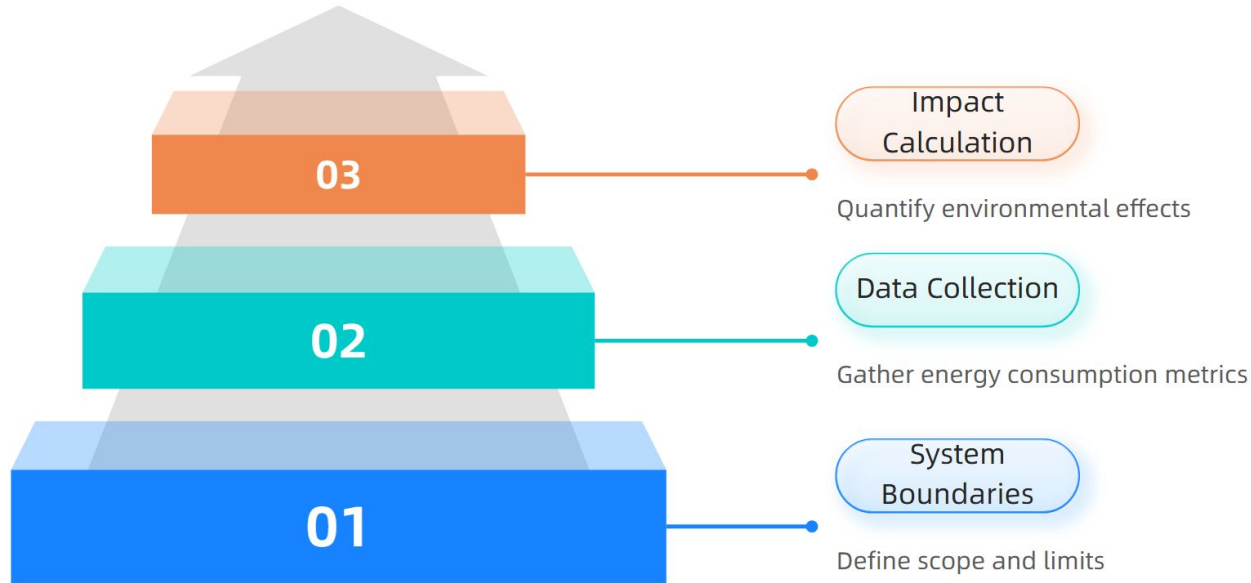


Scope Assessment

Direct and indirect emissions

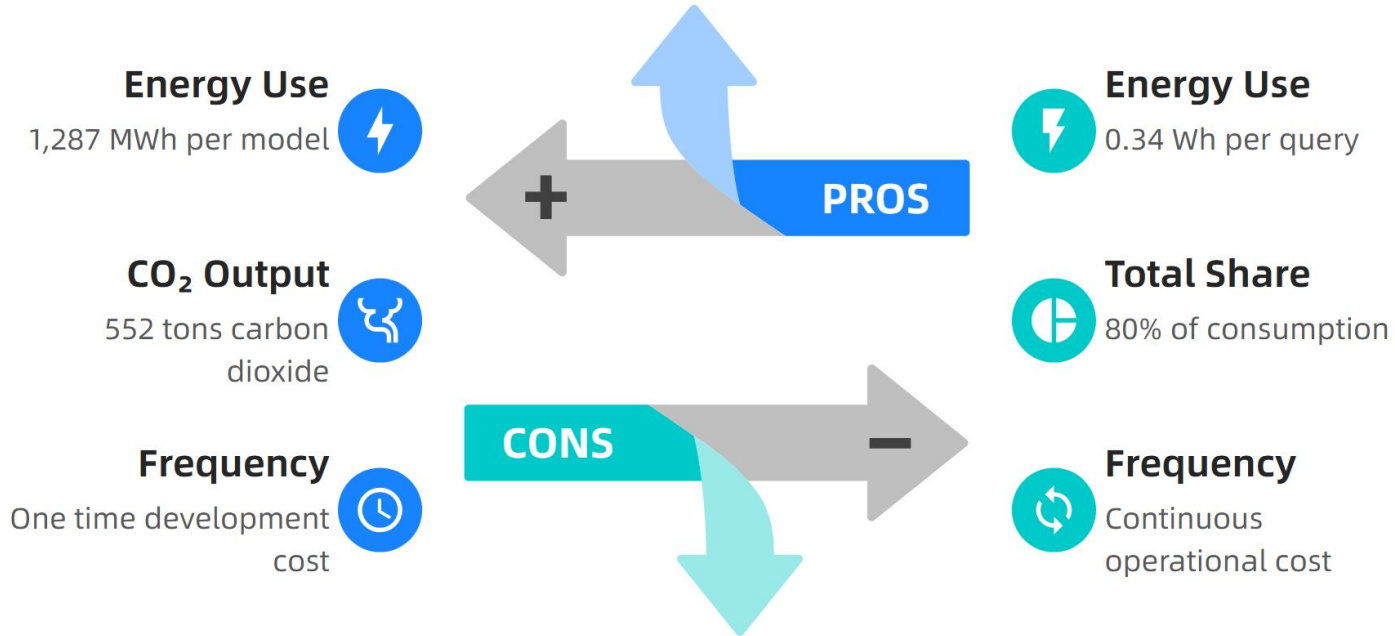


Environmental Impact Assessment Process



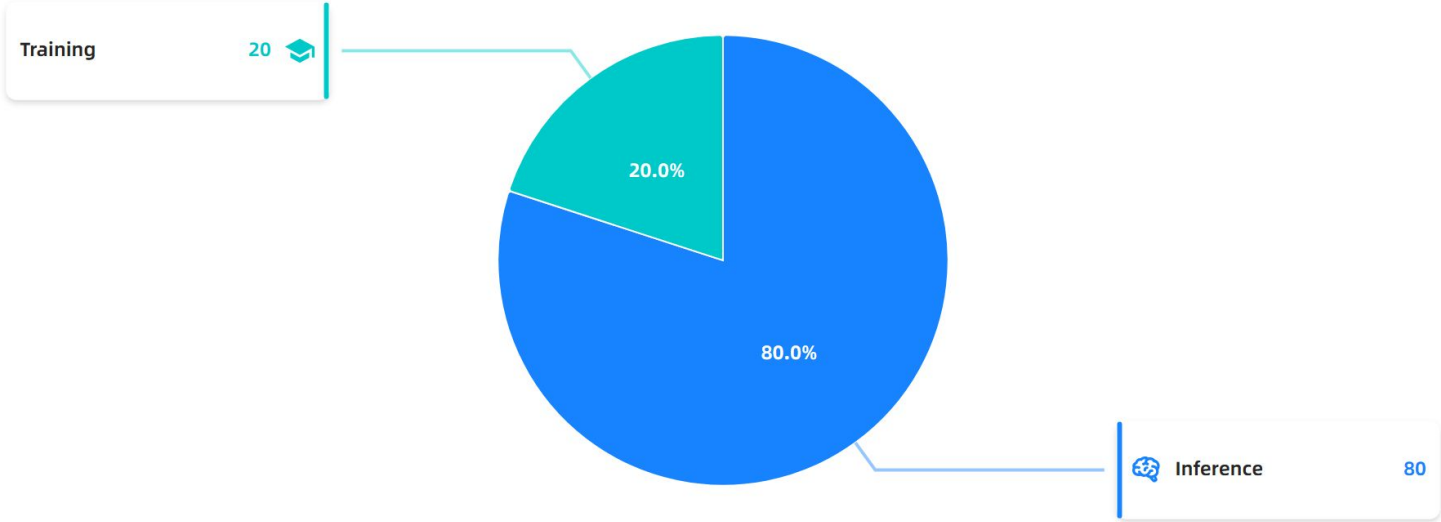
Source: Greenhouse Gas Protocol, 2024

Training vs Inference Comparison



Source: MIT Technology Review, 2025

AI Energy Consumption Distribution



Source: MIT Technology Review, 2025

Scope 2

Scope 3

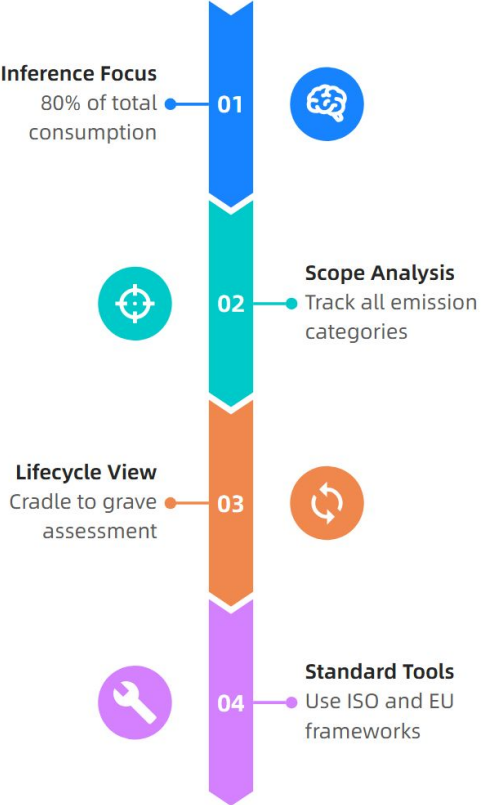
Scope 1

Assessment Frameworks Hierarchy



Source: *Environmental Footprint Methods*, European Commission, 2024

Key Assessment Takeaways



Source: Multiple Assessment Frameworks, 2024

Activity: Assess environmental impact (e.g., estimate energy footprint)

- Activity: AI Carbon Footprint Calculator
- Scenario: Your organization deploys a customer service chatbot handling 10,000 daily queries. Calculate the annual environmental impact using provided energy metrics and assessment frameworks.
- Step 1: Define system boundaries (training vs inference)
- Step 2: Calculate daily energy consumption (queries $\times$ 0.34 Wh)
- Step 3: Apply carbon intensity factors by region
- Step 4: Categorize emissions by GHG Protocol scopes
- Step 5: Generate environmental impact report
- Expected Output: Complete carbon footprint assessment with annual CO₂ estimates, energy consumption breakdown, and scope categorization

LO3 | LU3 | T1: Ethical frameworks and guidelines for generative AI-powered tools (including legal requirements for data protection)

What is Ethical frameworks and guidelines for generative AI-powered tools (including legal requirements for data protection)?



Legal Compliance
GDPR and regional privacy regulations



Data Protection
Privacy rights throughout AI lifecycle

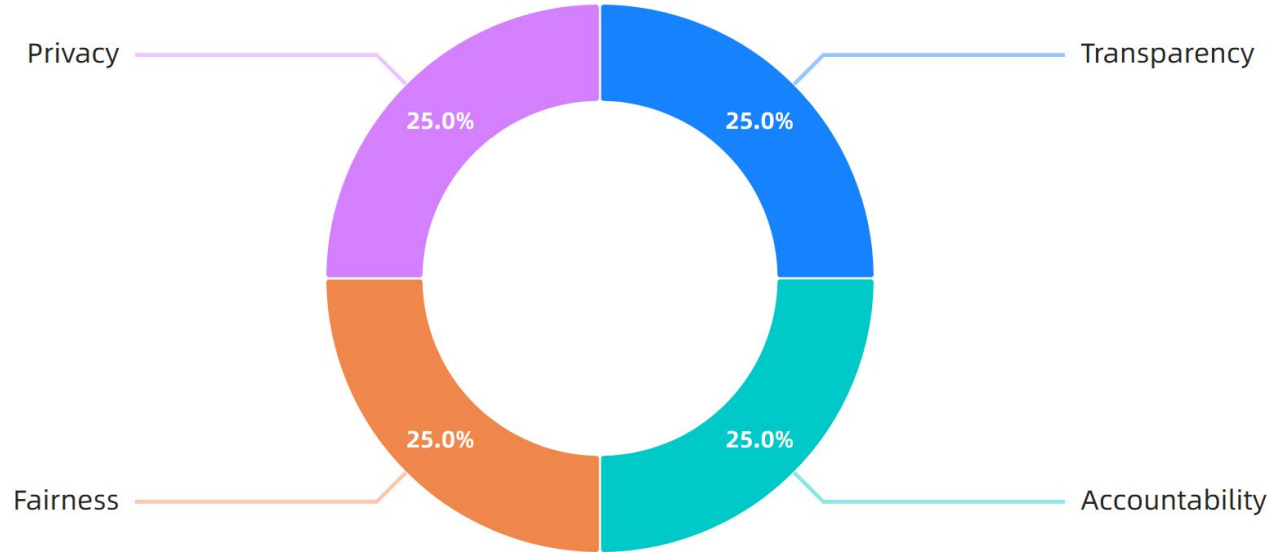


Governance Structure
AI Ethical Review Board oversight



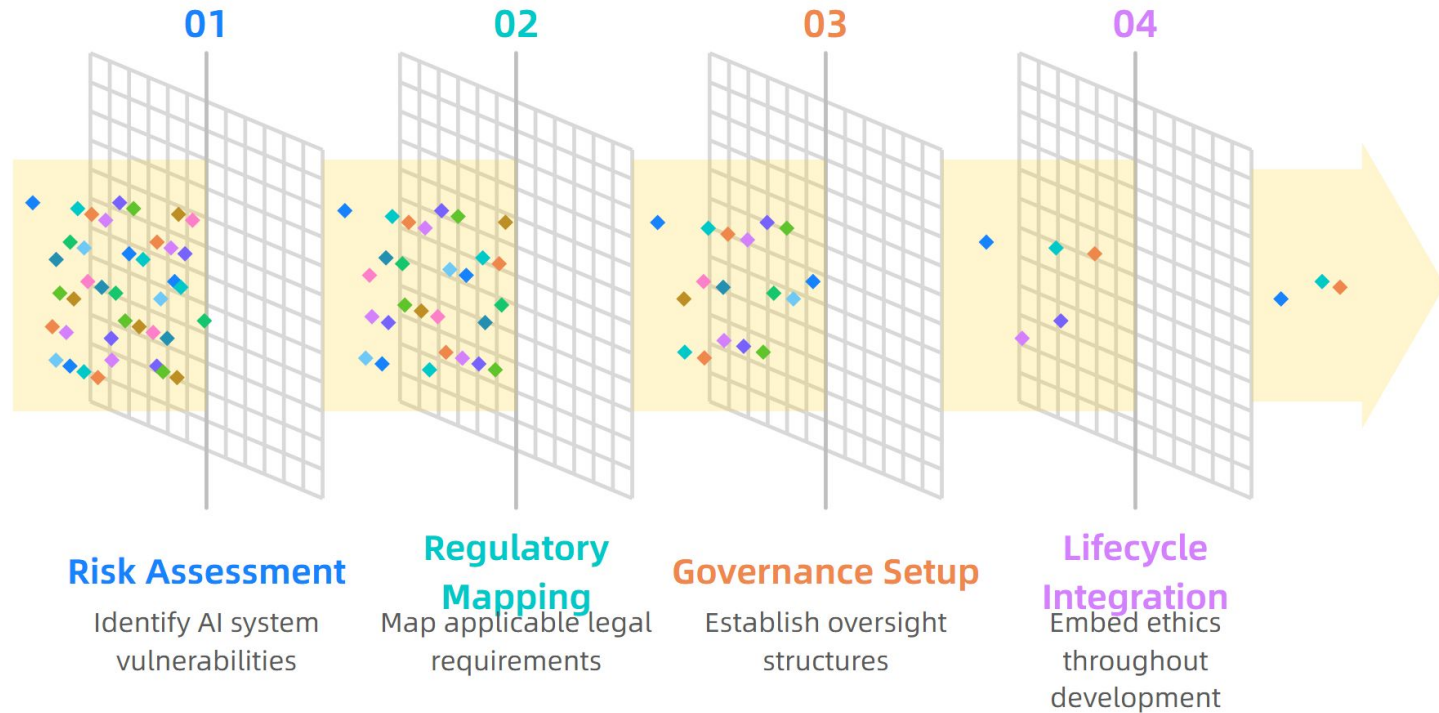
Industry Guidelines
Sector-specific ethical standards

Core ethical principles for AI implementation



Source: Harvard DCE, 2024

Step-by-step compliance framework implementation



Source: TrustArc AI Compliance, 2024

Traditional AI vs Generative AI ethical considerations

Limited Data

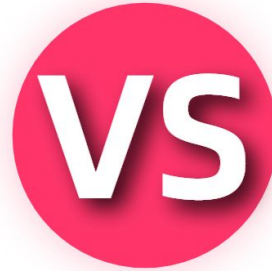
Structured datasets only

Predictable Outputs

Rule-based deterministic results

Lower Privacy Risk

Minimal personal data exposure



Large Datasets

Massive training data requirements

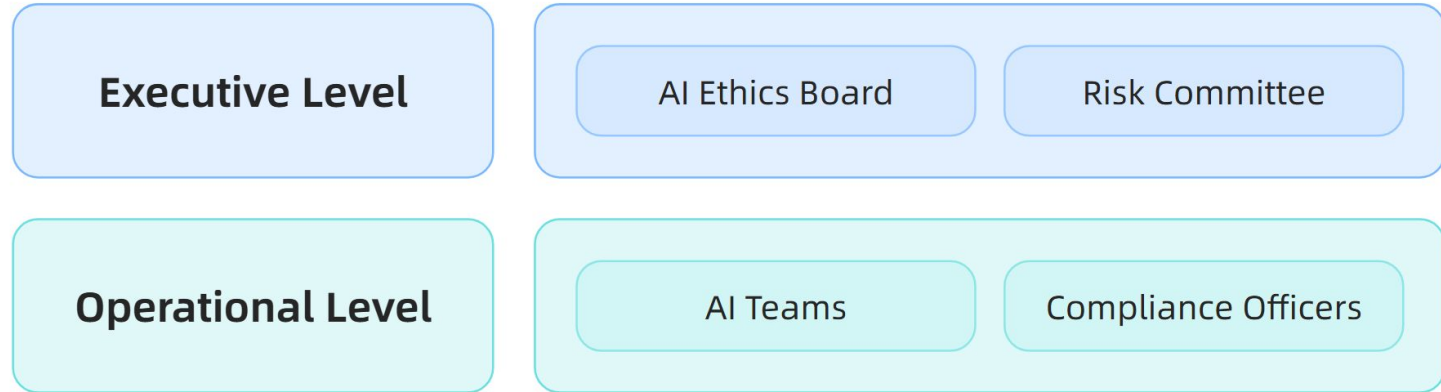
Complex Outputs

Unpredictable creative generation

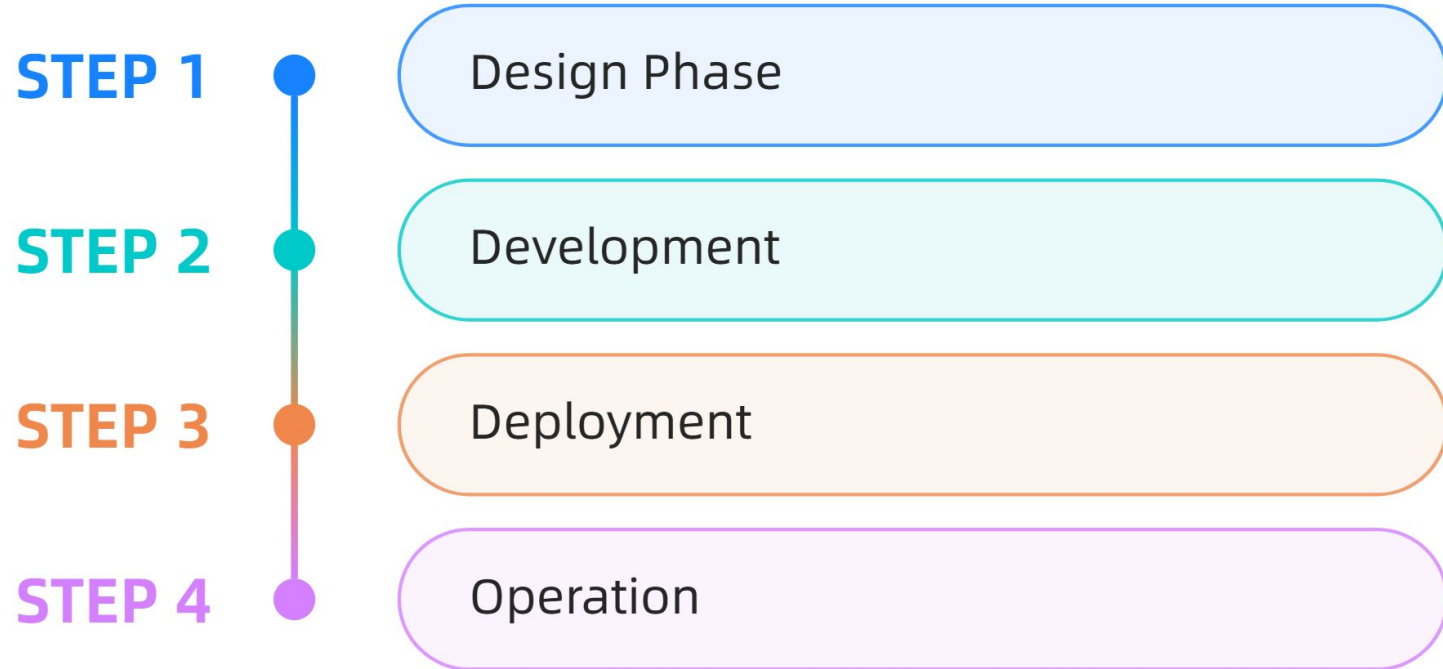
Higher Privacy Risk

Potential data memorization issues

AI governance hierarchy and oversight structure



Data protection timeline for AI systems



Key takeaways for implementing ethical AI frameworks

Human-Centric

Rights-based approach to
AI ethics

Lifecycle Integration

Ethics embedded
throughout development

Clear Accountability

Defined roles and
responsibilities

Continuous Monitoring

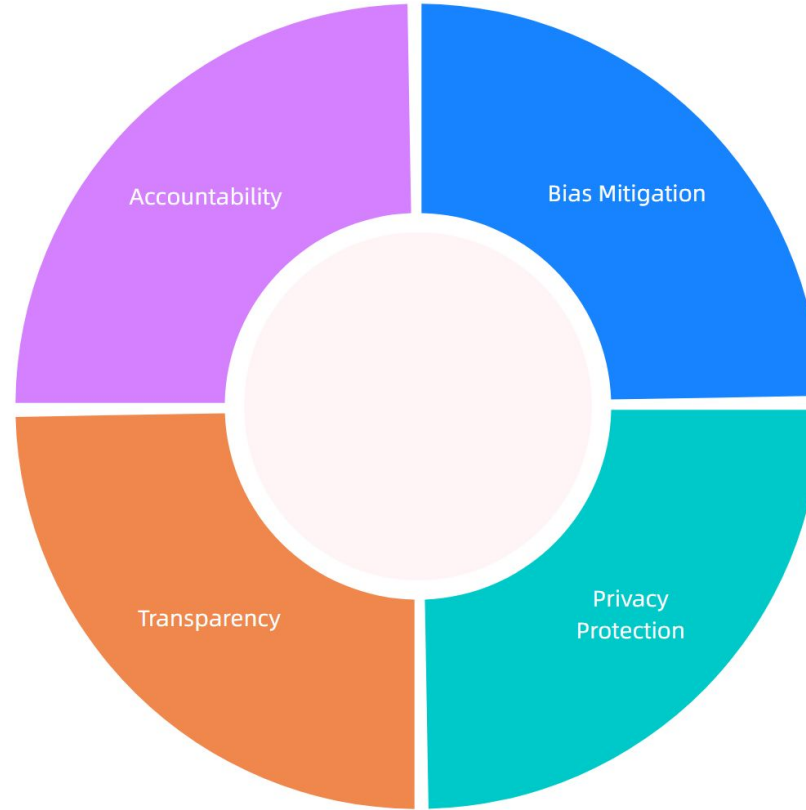
Ongoing compliance and
review

Activity: Ethical frameworks and guidelines for generative AI-powered tools (including legal requirements for data protection)

- Activity: AI Ethics Compliance Assessment
- Scenario: Your organization is implementing a generative AI chatbot for customer service. Conduct an ethical framework assessment to ensure compliance with data protection regulations.
- Step 1: Identify applicable data protection laws (GDPR, regional privacy acts)
- Step 2: Map ethical principles to chatbot functionality (transparency, privacy, fairness)
- Step 3: Design governance oversight structure with defined accountability roles
- Step 4: Create data minimization and encryption protocols for customer interactions
- Expected Output: Complete ethical compliance checklist with risk mitigation strategies
- Duration: 25 minutes

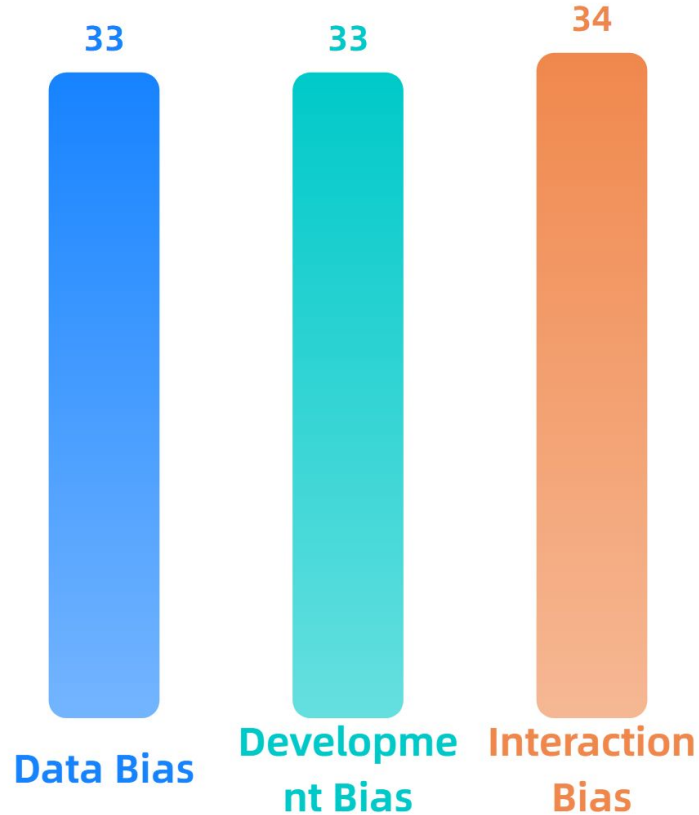
**LO3 | LU3 | T2: Ethical considerations in
AI development, (e.g., bias, privacy,
transparency, accountability)**

What are Ethical considerations in AI development, (e.g., bias, privacy, transparency, accountability)?



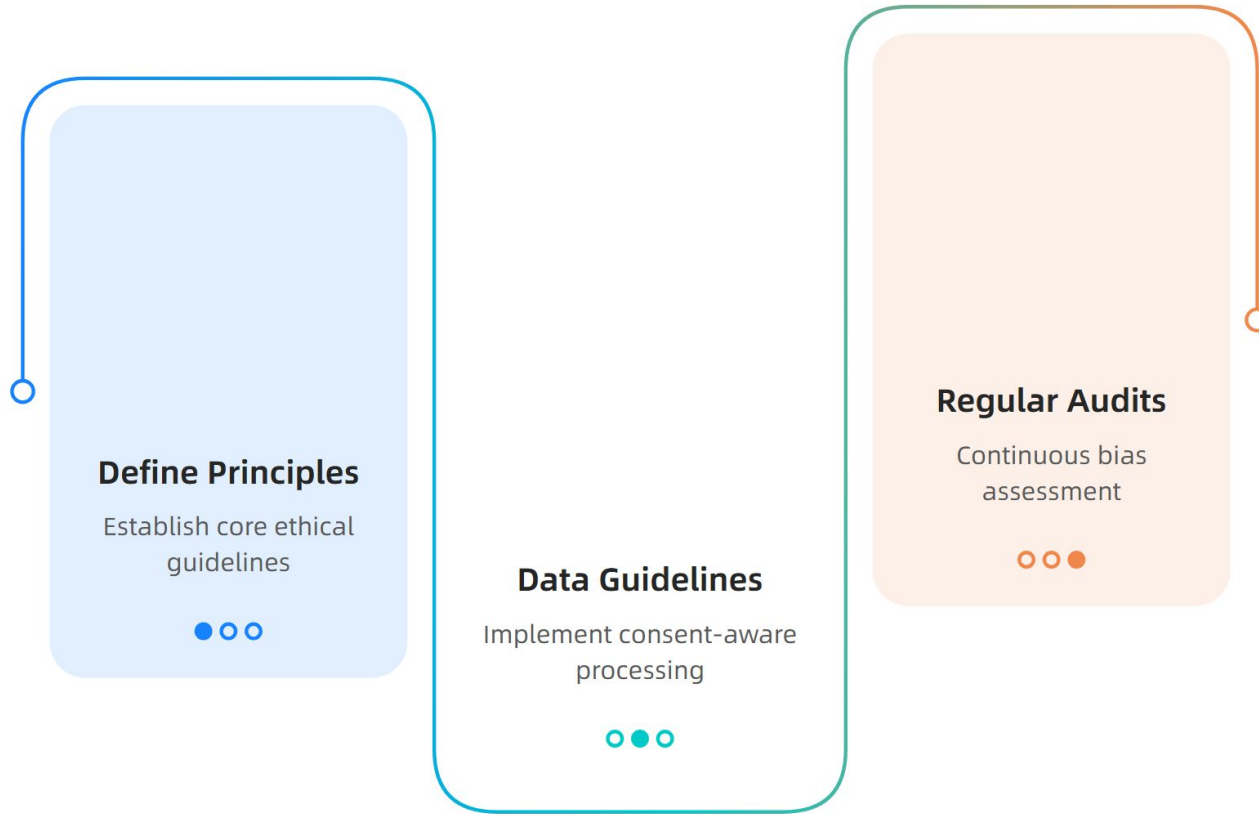
Source: UNESCO AI Ethics Recommendation, 2021

Types of AI Bias



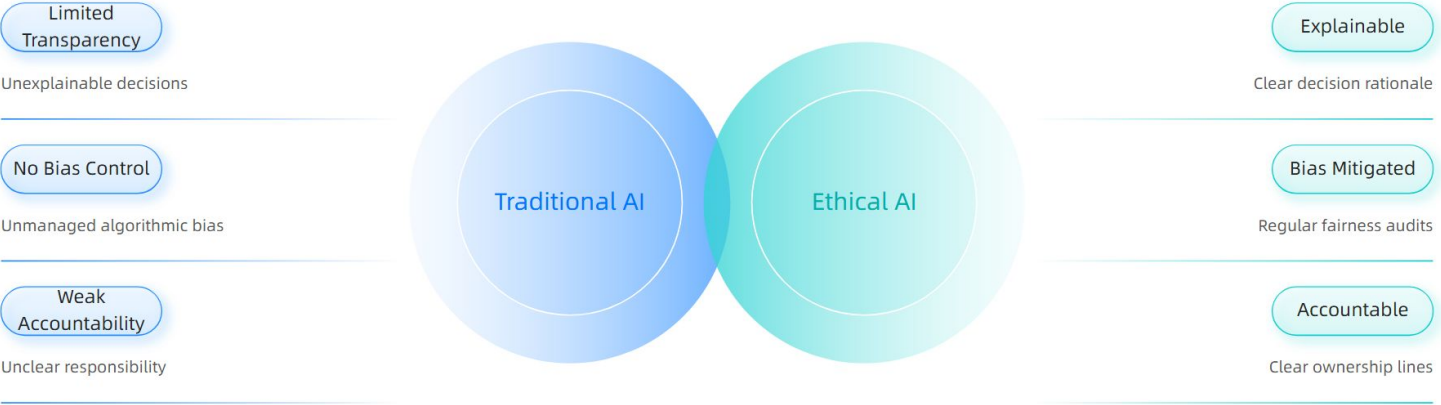
Source: PMC Research, 2024

Ethical AI Implementation Process



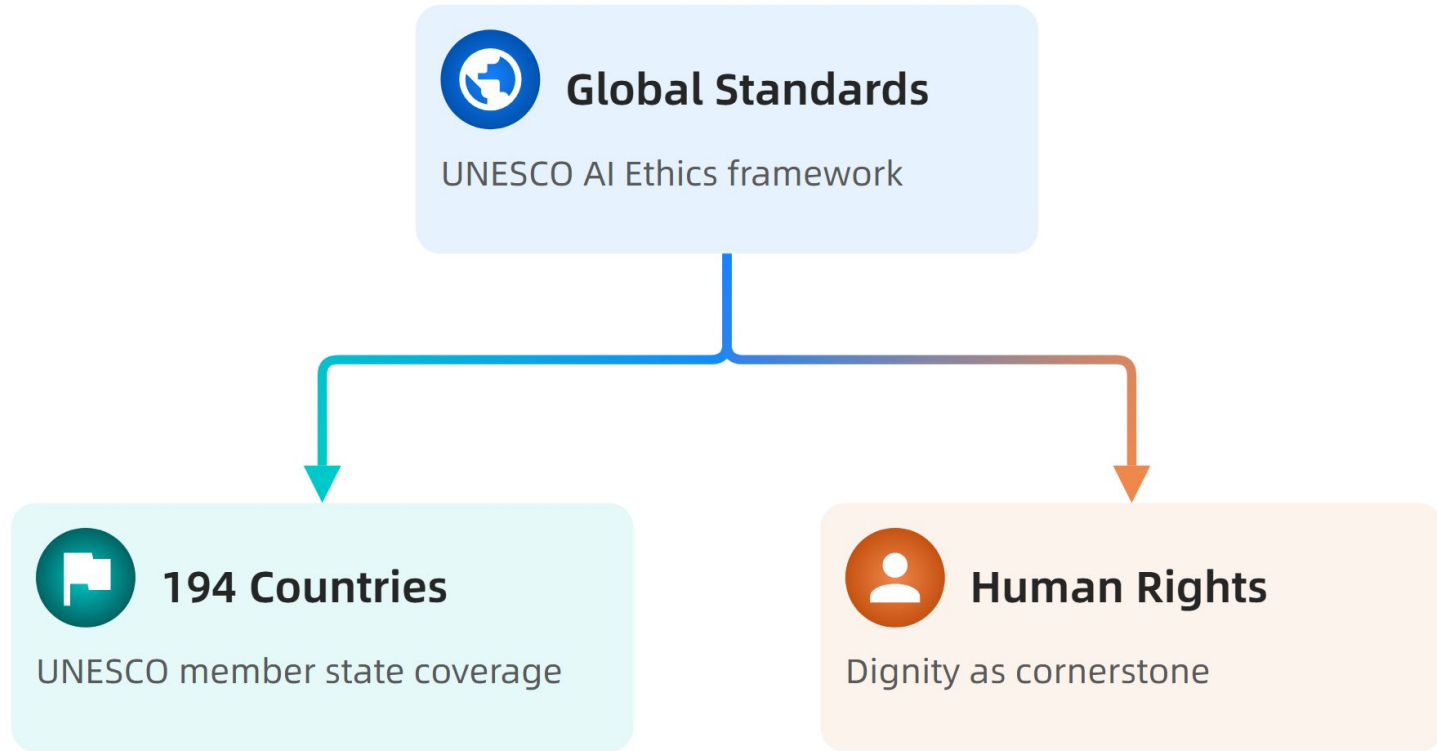
Source: IBM AI Ethics Guide, 2024

Traditional vs Ethical AI Systems

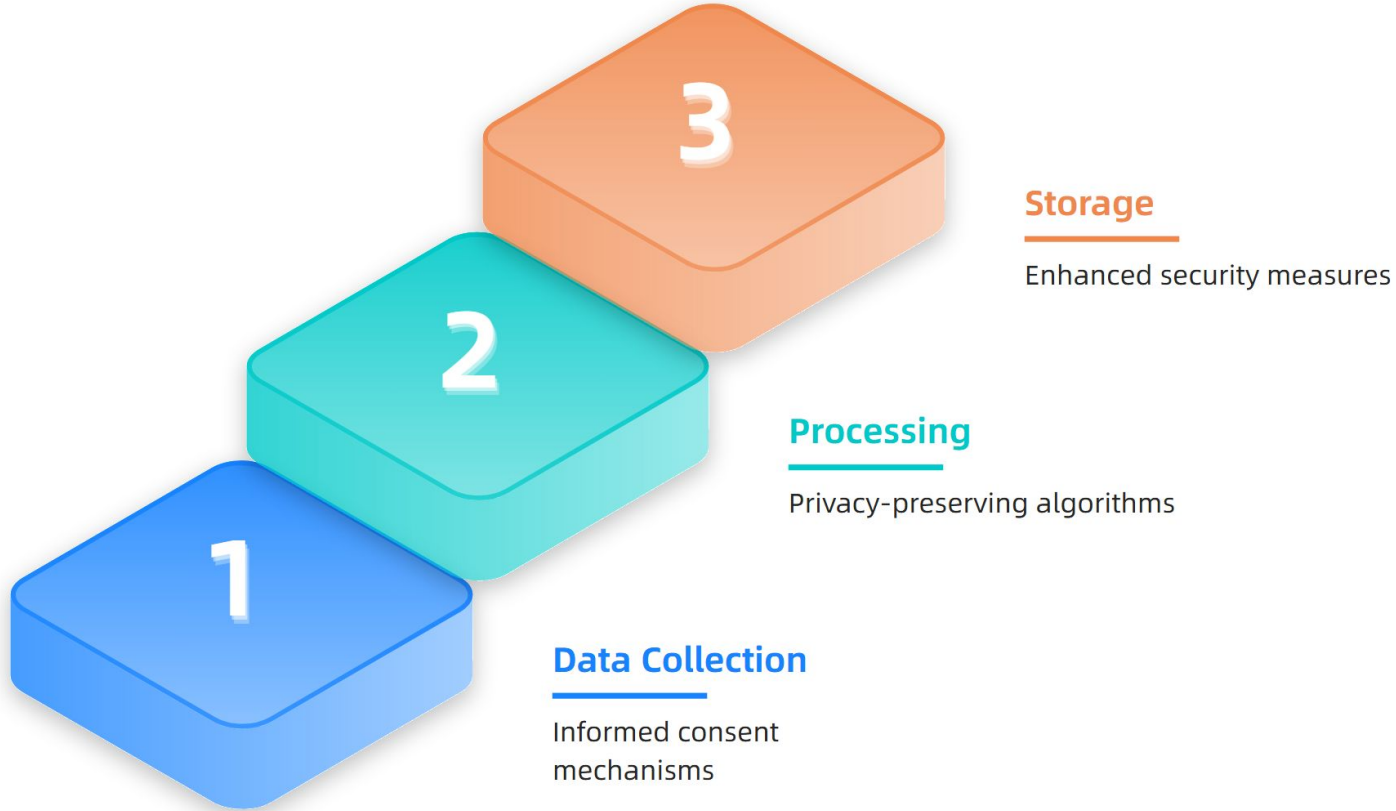


Source: Infosys BPM Research, 2024

Global Regulatory Framework



Privacy Protection Methods



Source: AI Ethics Integration Study, 2025

Key Takeaways for Ethical AI



Source: Multiple sources synthesis, 2024

Activity: Ethical considerations in AI development, (e.g., bias, privacy, transparency, accountability)

- Activity: AI Ethics Audit Exercise
- Scenario: Your organization is implementing a new AI-powered recruitment system. You must conduct an ethics assessment before deployment.
- Step 1: Identify potential bias sources in recruitment data
- Step 2: Design transparency measures for candidate feedback
- Step 3: Establish accountability framework for hiring decisions
- Step 4: Create privacy protection plan for candidate data
- Expected Output: Complete ethics assessment report with mitigation strategies
- Duration: 25 minutes

**LO3 | LU3 | T3: Develop strategies to
integrate ethics into AI development and
evaluation of AI systems**

What is Develop strategies to integrate ethics into AI development and evaluation of AI svstems?



Ethics Design

Principles integrated throughout AI lifecycle



Evaluation Tools

Systematic measurement of ethical compliance



Stakeholder Input

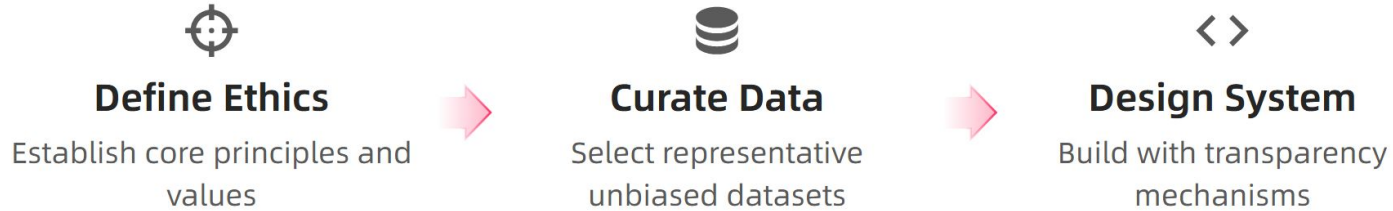
Diverse community engagement in decisions



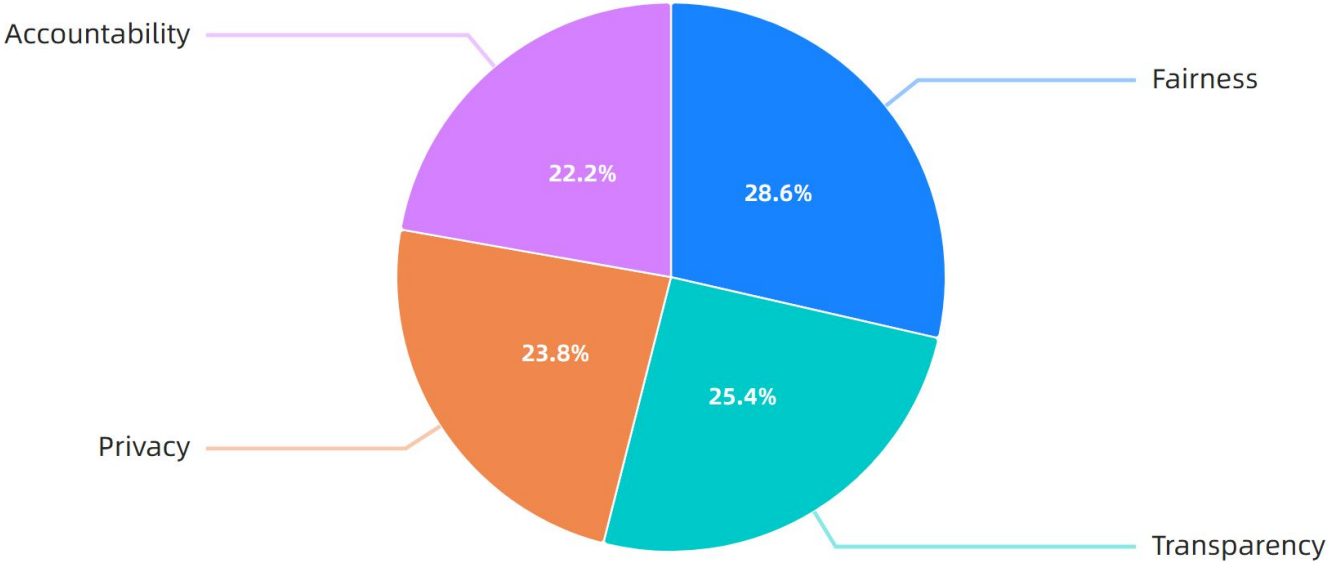
Governance Frame

Oversight mechanisms for accountability

Ethics-by-Design Implementation Process



Ethical Principles Distribution in AI Systems



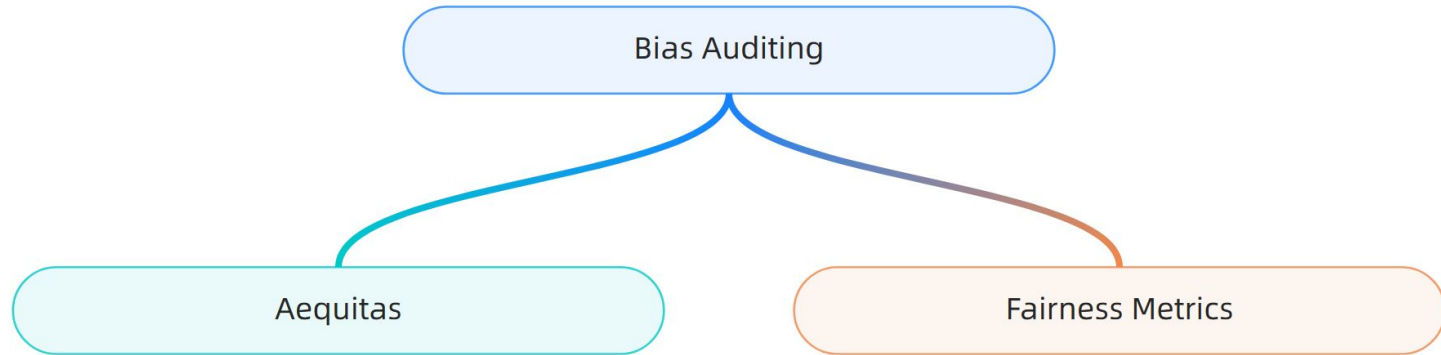
Source: Nature Scientific Data, 2024 (12,067 data points)

Traditional vs Ethics-by-Design Approaches

T TRADITIONAL DEV	E ETHICS DESIGN
● Ethics Afterthought ● Compliance Focus ● Limited Input	● Integrated Throughout ● Risk Management ● Diverse Engagement

Source: EU Digital Strategy, 2024

Practical Evaluation Tools and Frameworks



Source: Top Metrics for Evaluating Ethical AI, 2024

Evolution of Ethical AI Frameworks



Source: Responsible AI Research spanning 2011-2023

Key Takeaways for Ethical AI Strategy



Lifecycle Integration



Measurable Metrics



Cross Function



Continuous Monitor

Activity: Develop strategies to integrate ethics into AI development and evaluation of AI systems

- Activity: Ethics Integration Assessment
- Scenario: Your organization is developing a customer service chatbot that handles sensitive personal data and makes decisions about service eligibility.
- Step 1: Identify three ethical principles most relevant to this AI system
- Step 2: Select appropriate evaluation tools from the toolkit for bias and transparency
- Step 3: Design a stakeholder engagement plan including affected customer groups
- Step 4: Create a monitoring framework with specific metrics for ongoing compliance
- Expected Output: Complete ethical integration plan with principles, tools, stakeholders, and metrics
- Duration: 25 minutes

LO3 | LU3 | T4: Apply ethical guidelines to AI development and use, considering fairness, accountability, transparency and sustainability etc.

What is Apply ethical guidelines to AI development and use, considering fairness, accountability, transparency and sustainability etc.?

Fairness



Eliminate bias across all populations

Accountability



Clear responsibility for AI decisions

Transparency



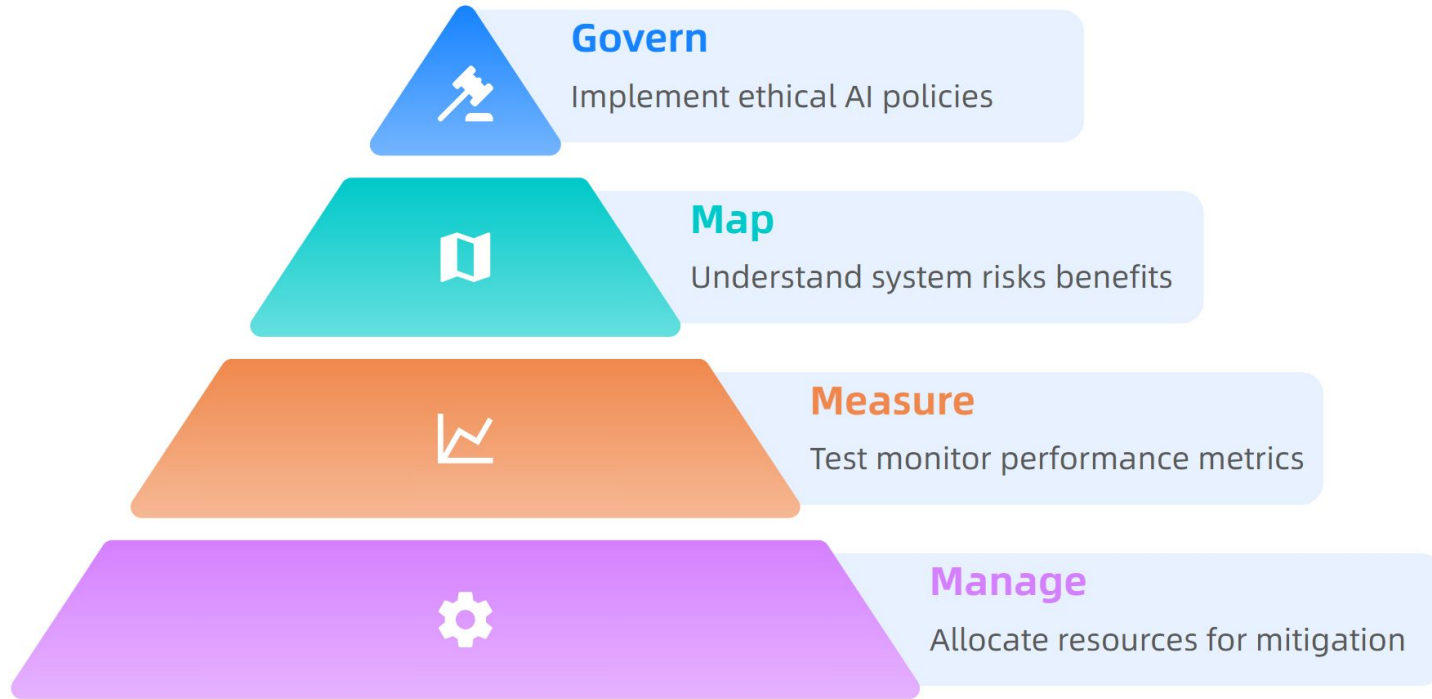
Open explainable AI processes

Sustainability



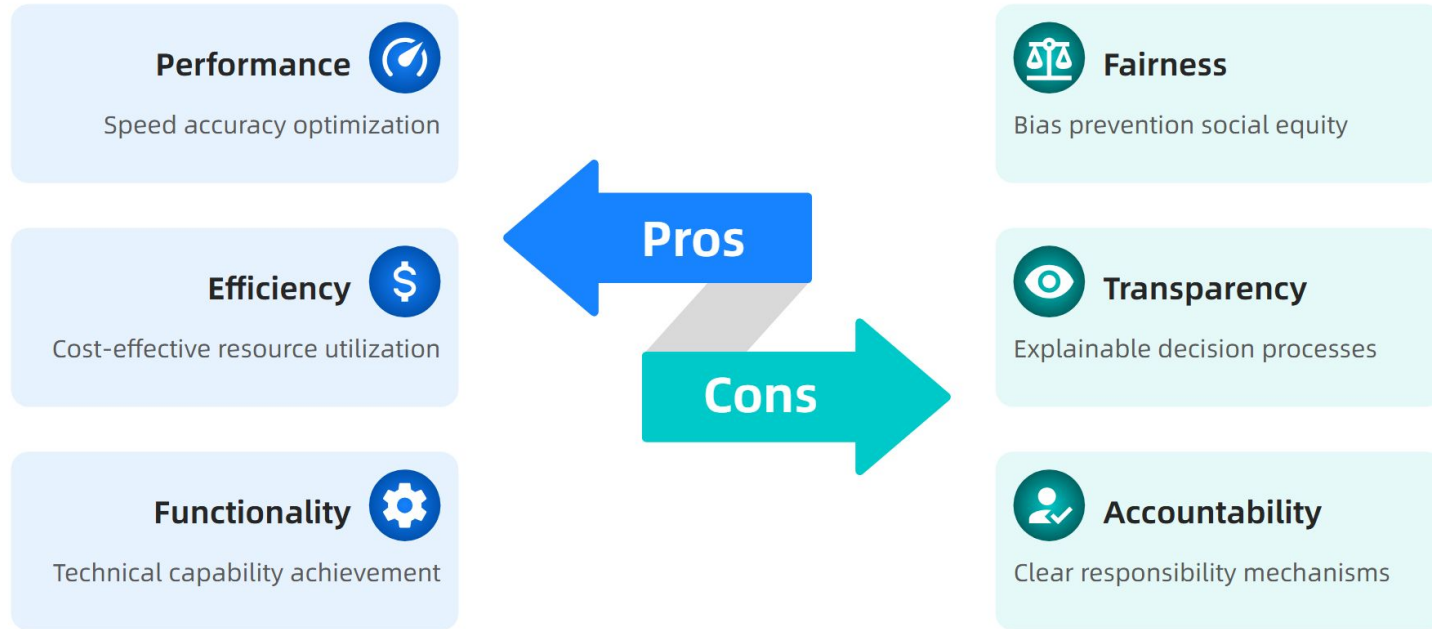
Environmental and social responsibility

NIST AI Risk Management Process



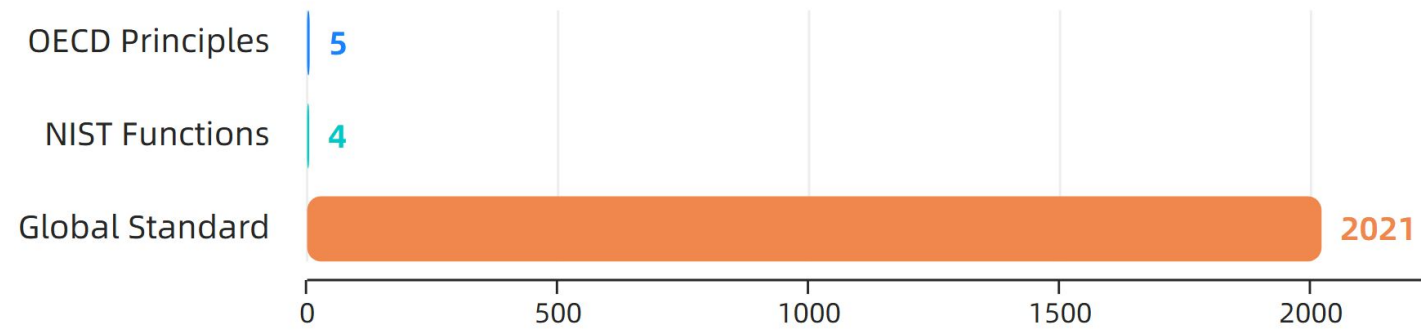
Source: NIST AI Risk Management Framework (Bradley, 2025)

Traditional vs Ethical AI Approaches



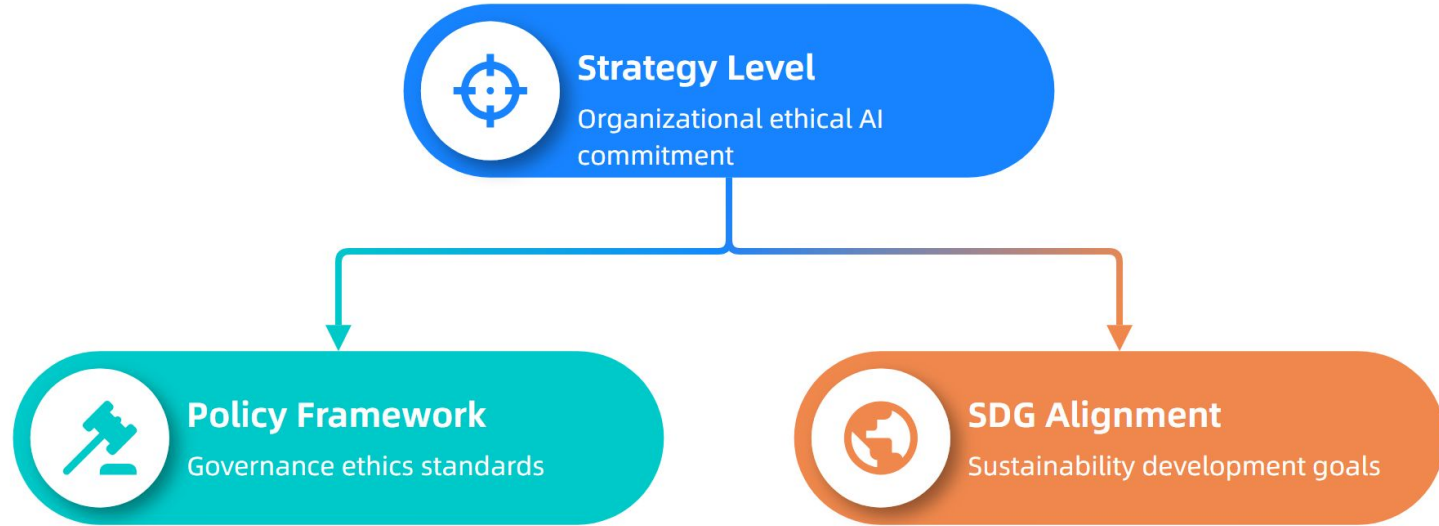
Source: Research synthesis from multiple frameworks

Global AI Ethics Framework Statistics



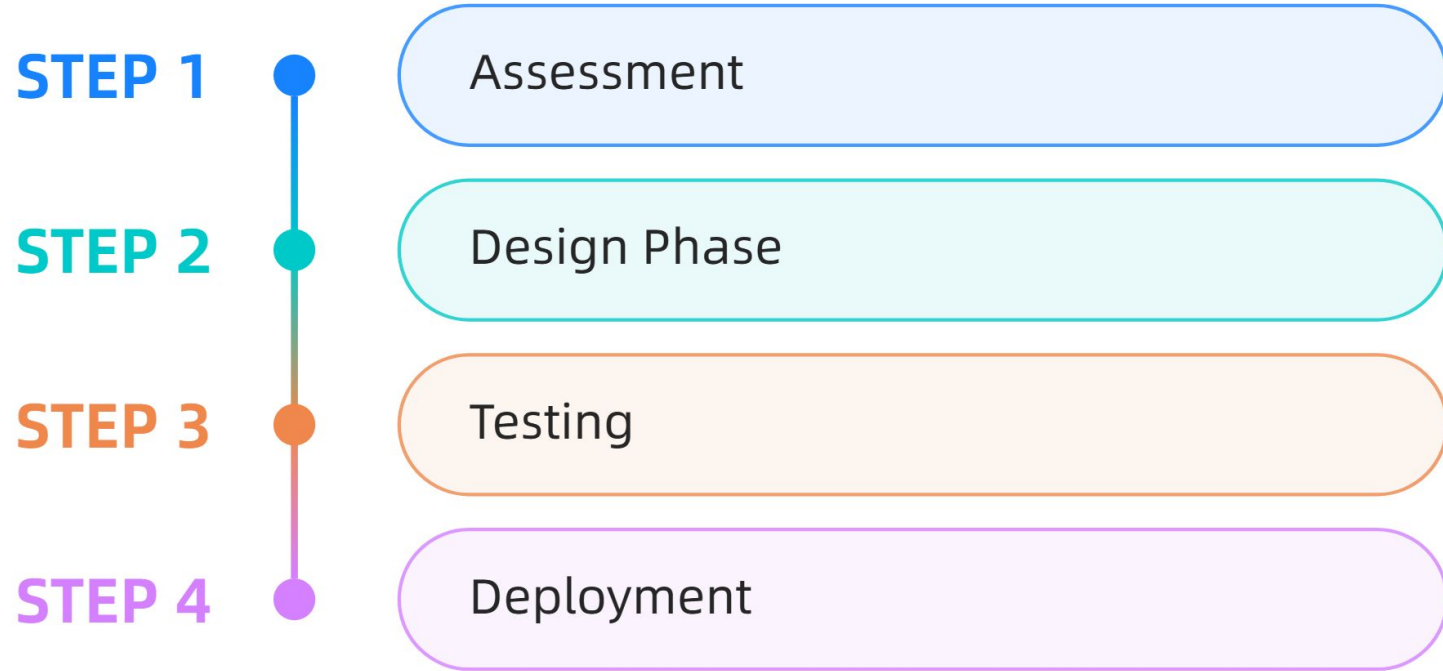
Source: OECD, NIST, UNESCO frameworks (Bradley, 2025)

Implementation Strategy Hierarchy



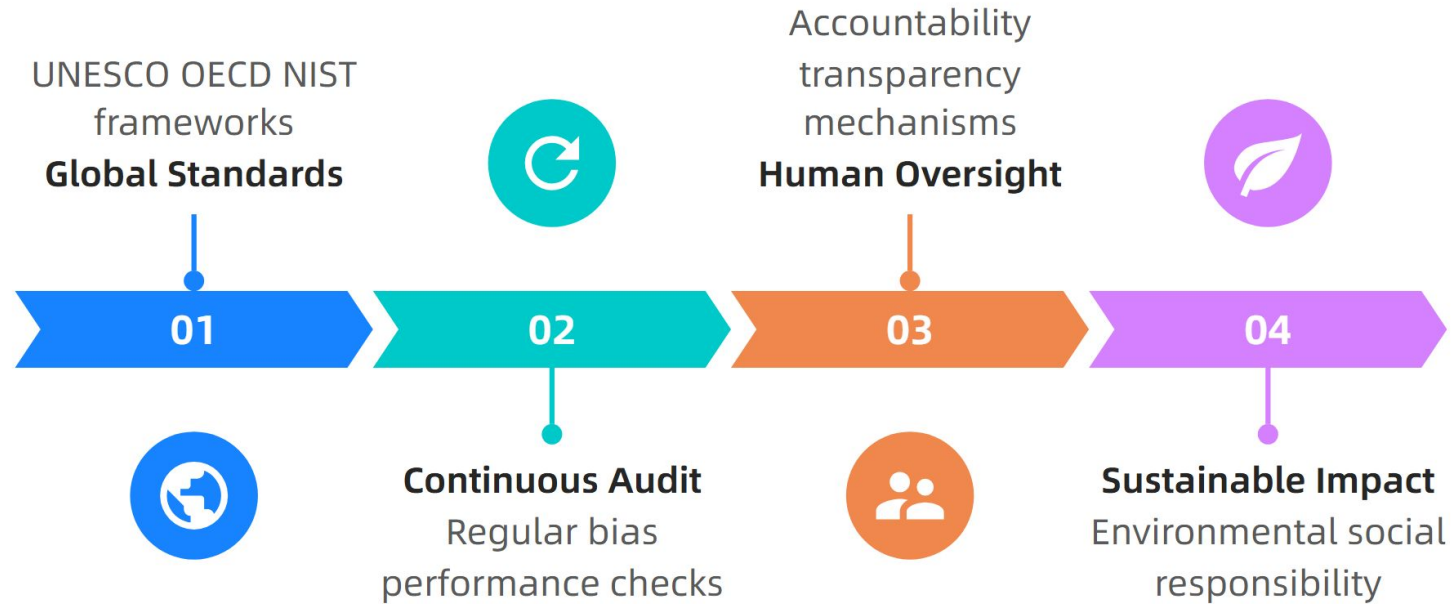
Source: UNESCO Recommendation and ALTAI (UNESCO, 2021)

Ethical AI Implementation Timeline



Source: *ALTAI Guidelines and IEEE Standards (EU, 2024)*

Key Takeaways for Ethical AI



Source: Multi-framework synthesis (UNESCO, OECD, NIST)

Activity: Apply ethical guidelines to AI development and use, considering fairness, accountability, transparency and sustainability etc.

- Activity: Ethical AI Assessment Workshop
- Scenario: Your organization is developing a recruitment AI system. Conduct an ethical assessment using ALTAI framework to identify potential bias risks and implementation strategies.
- Step 1: Review ALTAI checklist categories
- Step 2: Identify bias risks in recruitment data
- Step 3: Design transparency mechanisms for candidates
- Step 4: Create accountability framework for decisions
- Step 5: Present ethical implementation plan
- Expected Output: Comprehensive ethical assessment report with actionable recommendations
- Duration: 45 minutes

**LO4 | LU4 | T1: Trustworthy and
transparent communication of
generative model capabilities and
limitations**

What is Trustworthy and transparent communication of generative model capabilities and limitations?



Clear Documentation
Open model architecture and data sources



Limitation Awareness
Honest disclosure of model constraints

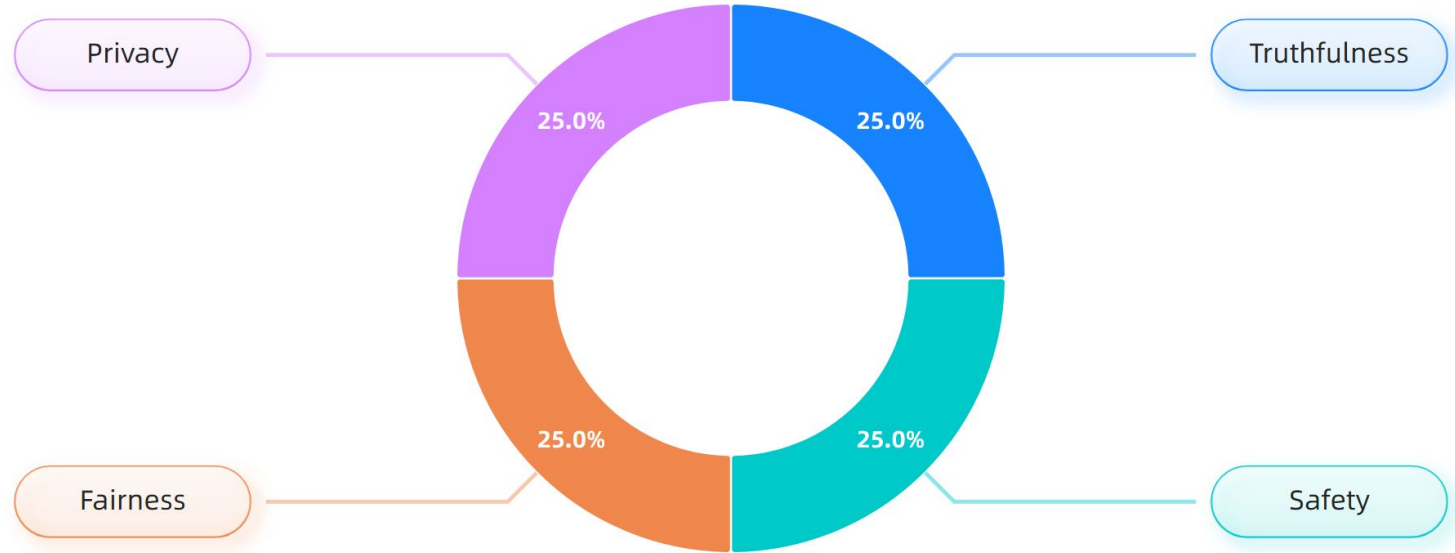


Stakeholder Trust
Building confidence through transparency



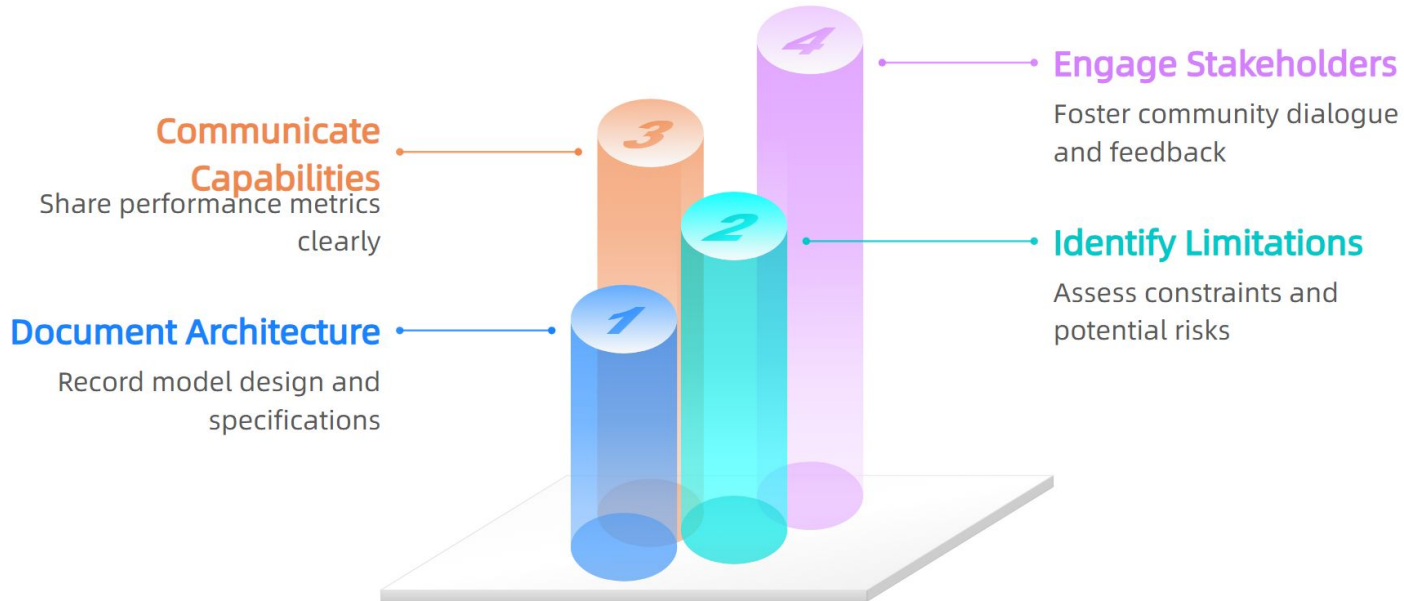
Ongoing Dialogue
Active community engagement and feedback

TrustGen Framework Dimensions



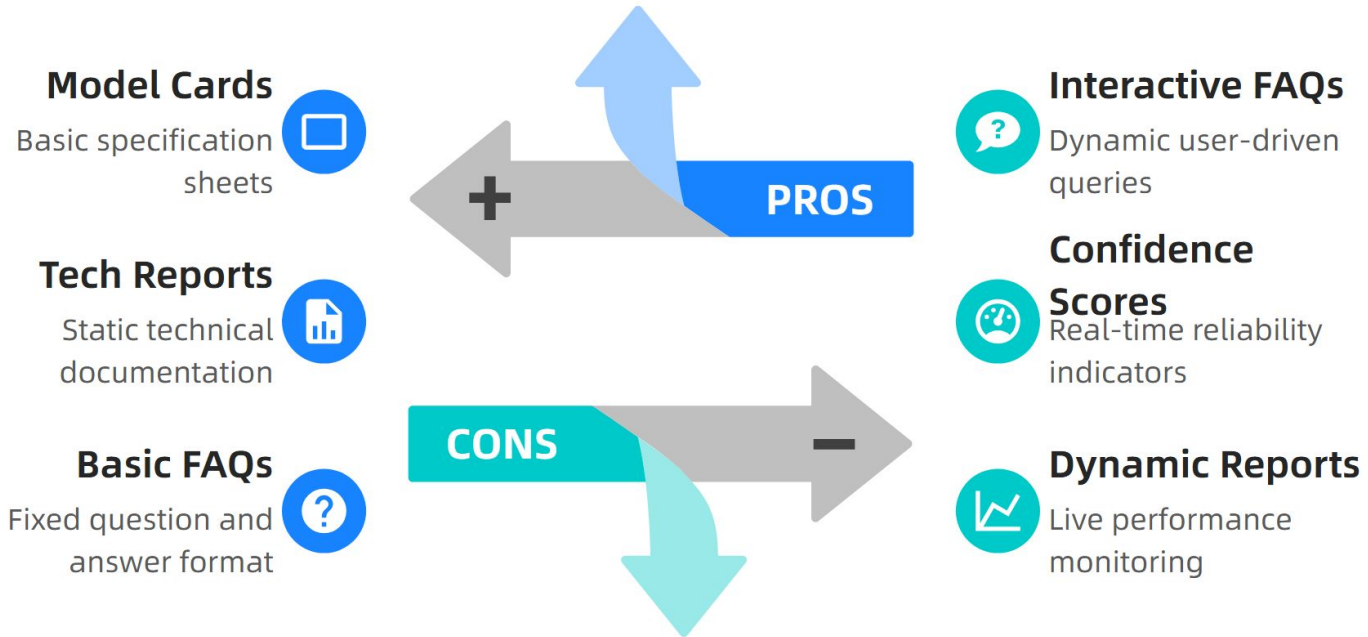
Source: TrustGen Framework, 2024

Implementation Process Steps



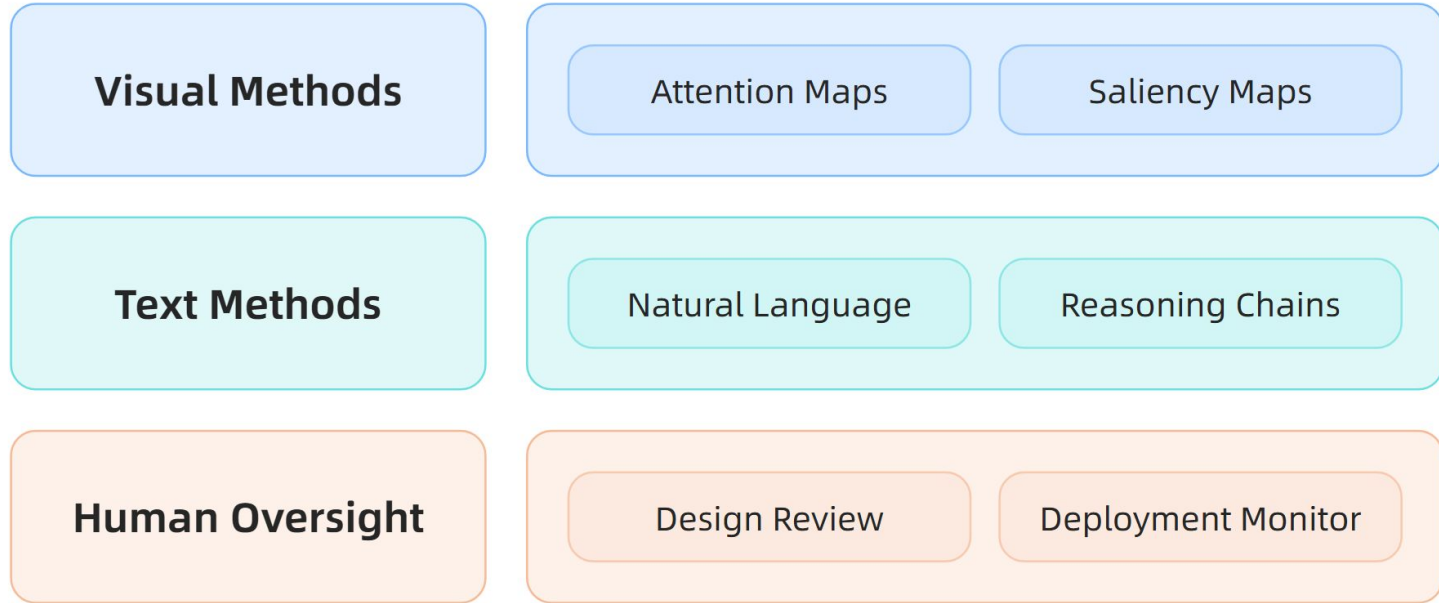
Source: *Building Trust in Generative AI*, 2024

Communication Approaches Comparison

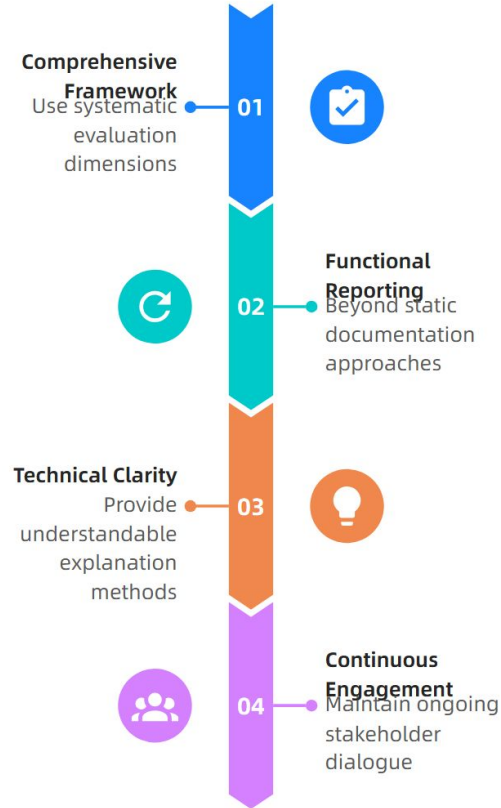


Source: MIT Press Human-Centered Research, 2024

Technical Explanation Methods



Key Takeaways for Implementation



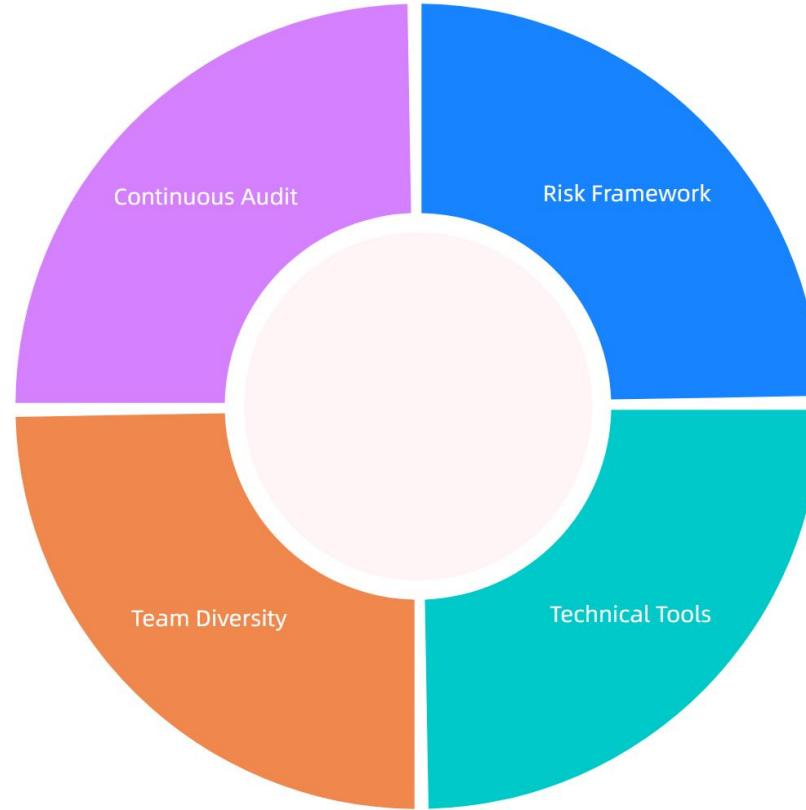
Source: Responsible AI Development Guidelines, 2024

Activity: Trustworthy and transparent communication of generative model capabilities and limitations

- Activity: Transparency Audit Exercise
- Scenario: Your organization is deploying a new customer service chatbot powered by a large language model. Stakeholders are concerned about AI transparency and want comprehensive documentation of capabilities and limitations before launch.
- Step 1: Create a TrustGen evaluation scorecard for your chatbot across all seven dimensions
- Step 2: Draft functional documentation including confidence indicators and interactive FAQ elements
- Step 3: Design stakeholder communication plan with specific transparency commitments and feedback channels
- Step 4: Present your transparency framework to the class for peer review and improvement suggestions
- Expected Output: Complete transparency audit report with evaluation metrics, documentation plan, and stakeholder engagement strategy

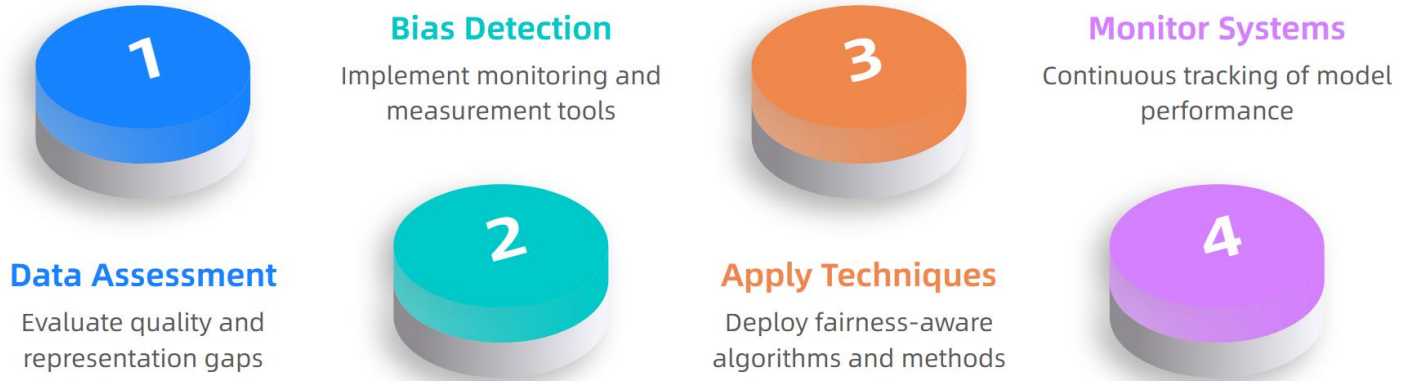
LO4 | LU4 | T2: Implement strategies to mitigate bias and ethical risks in AI systems

What is Implement strategies to mitigate bias and ethical risks in AI systems?



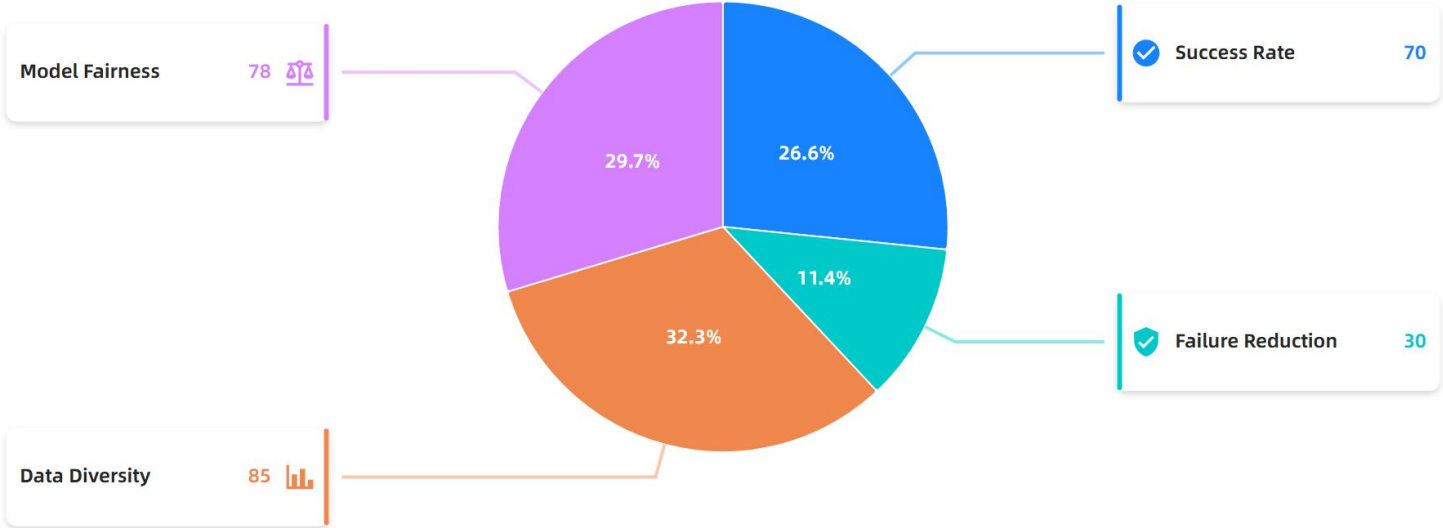
Source: NIST AI Risk Management Framework, 2024

Implementation Process



Source: *Fairness and Bias in AI Survey, 2023*

Success Metrics



Source: Algorithmic Bias Detection, Brookings 2024

Approach Comparison

Limited Detection

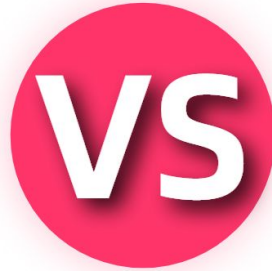
Basic bias identification
methods

Reactive Response

Address issues after
deployment

Single Metrics

Narrow performance
measurement focus



Proactive Mitigation

Preventive bias reduction
techniques

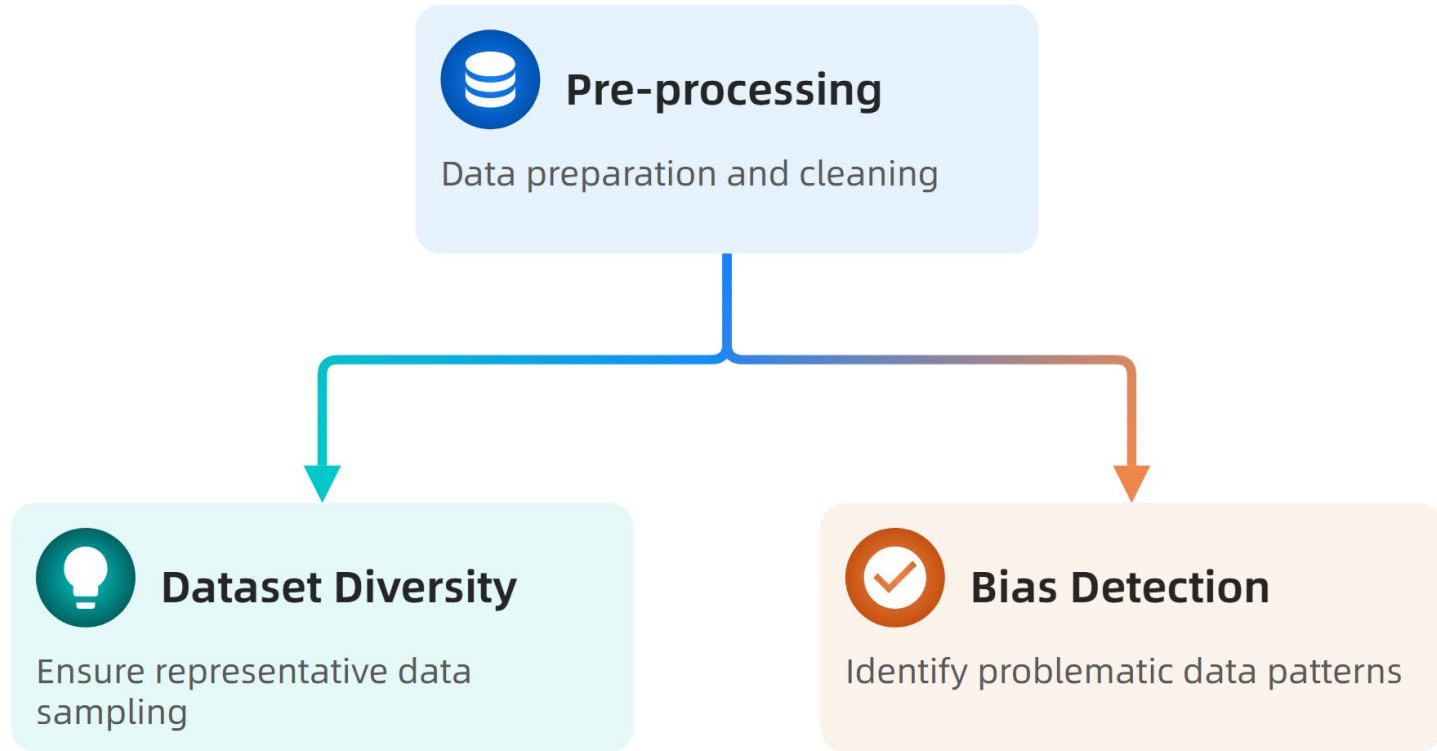
Continuous Monitoring

Ongoing system performance
tracking

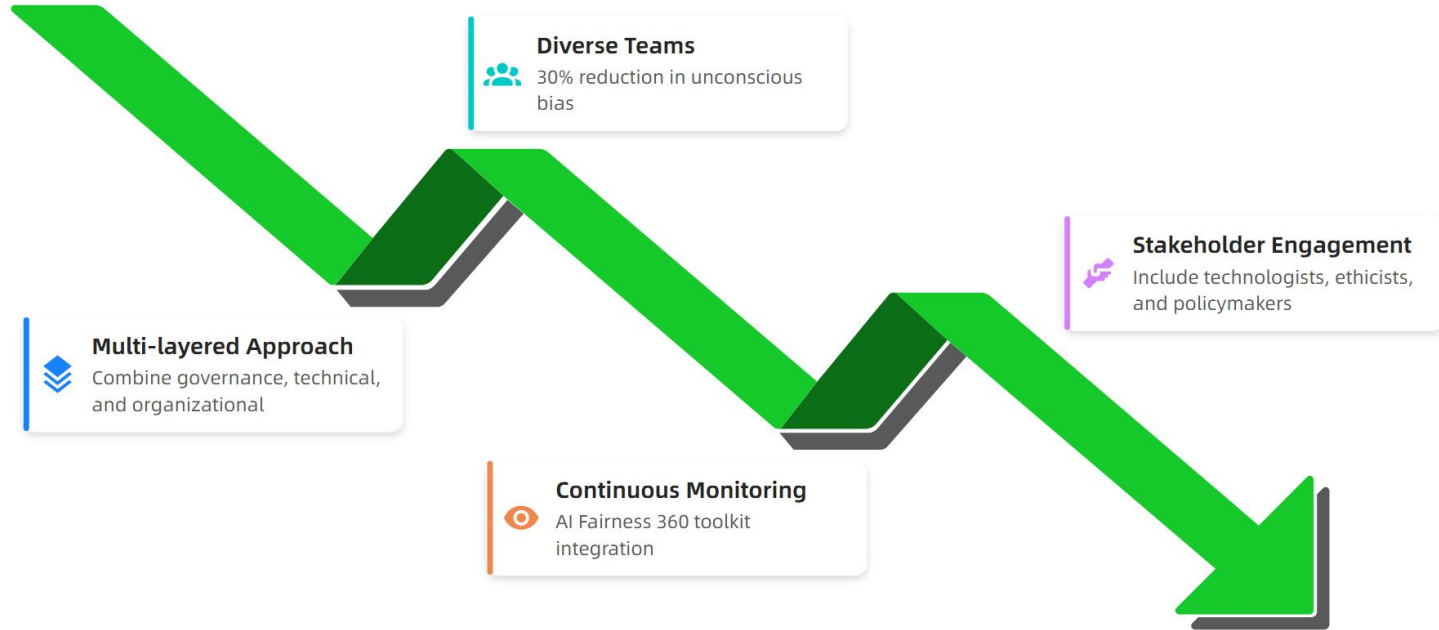
Multiple Metrics

Comprehensive fairness
evaluation

Technical Solutions



Key Takeaways



Source: Multiple sources integration, 2024

Activity: Implement strategies to mitigate bias and ethical risks in AI systems

- Activity: AI Ethics Board Simulation
- Scenario: Your organization is deploying a recruitment AI system. As the ethics board, evaluate the proposed bias mitigation strategy and identify gaps.
- Step 1: Review provided AI system documentation
- Step 2: Apply NIST AI RMF assessment criteria
- Step 3: Identify potential bias sources and risks
- Step 4: Recommend specific mitigation techniques
- Step 5: Design monitoring and audit procedures
- Expected Output: Comprehensive bias mitigation plan with specific techniques, metrics, and governance procedures

**LO4 | LU4 | T3: Champion best practices
for responsible and ethical use of
generative AI, addressing concerns
about transparency, bias, and societal
impact**

What is Champion best practices for responsible and ethical use of generative AI, addressing concerns about transparency, bias, and societal impact?



Transparency

Clear algorithmic and data processes



Bias Mitigation

Systematic fairness and audit techniques



Accountability

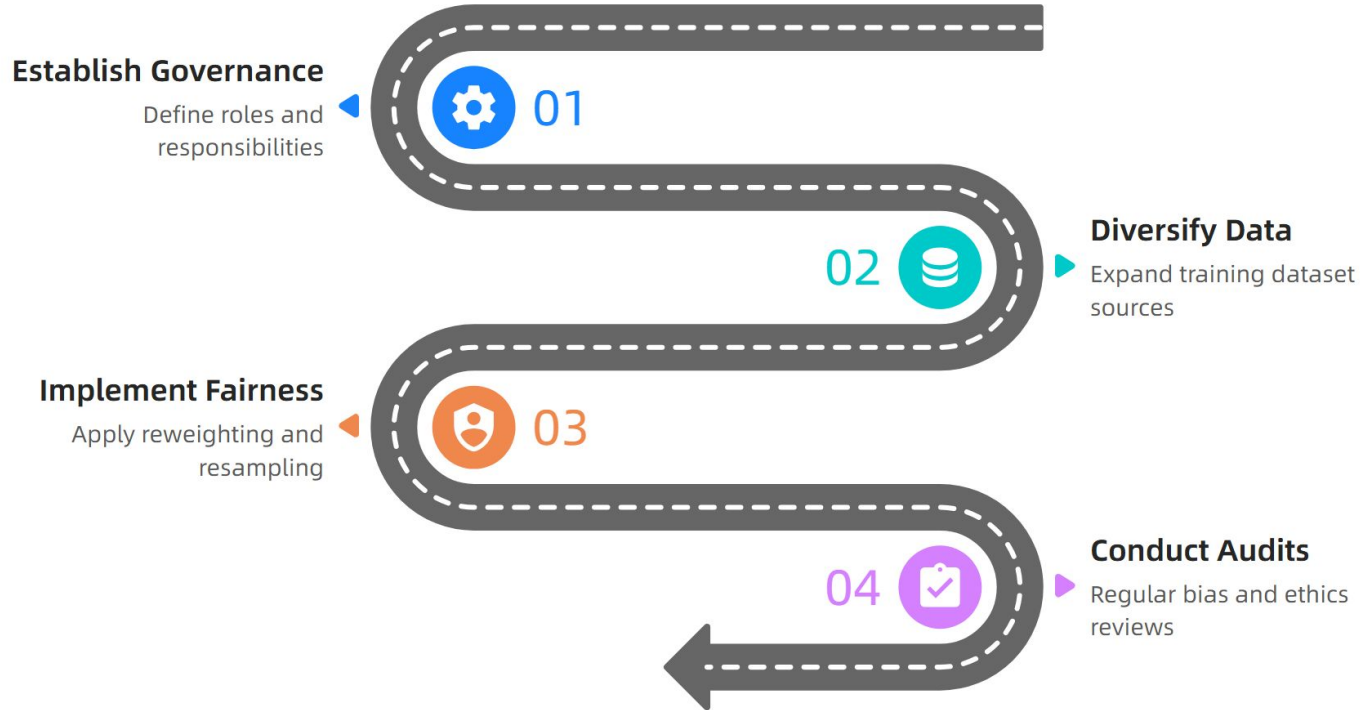
Governance frameworks and monitoring systems



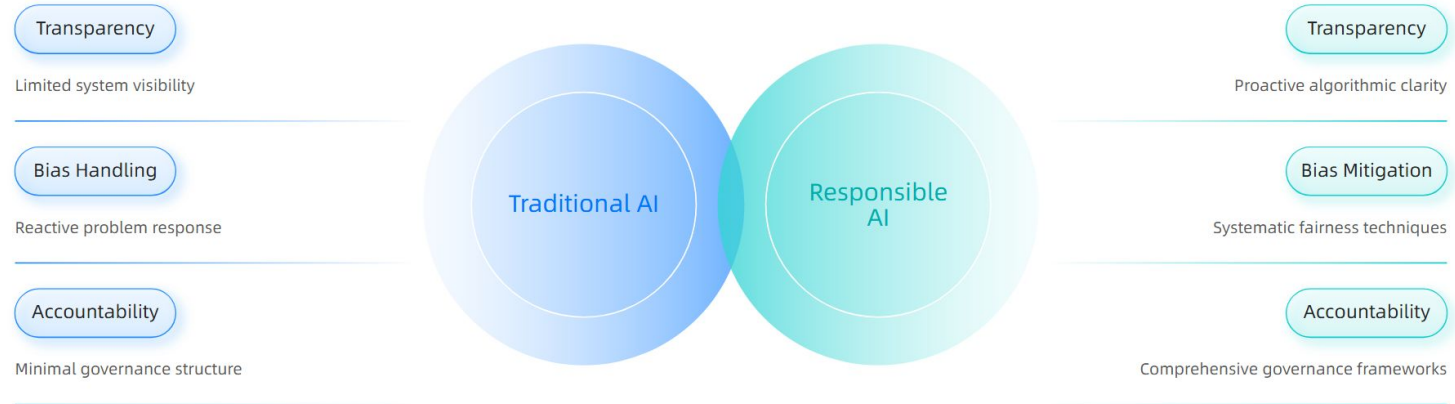
Societal Impact

Address public concerns proactively

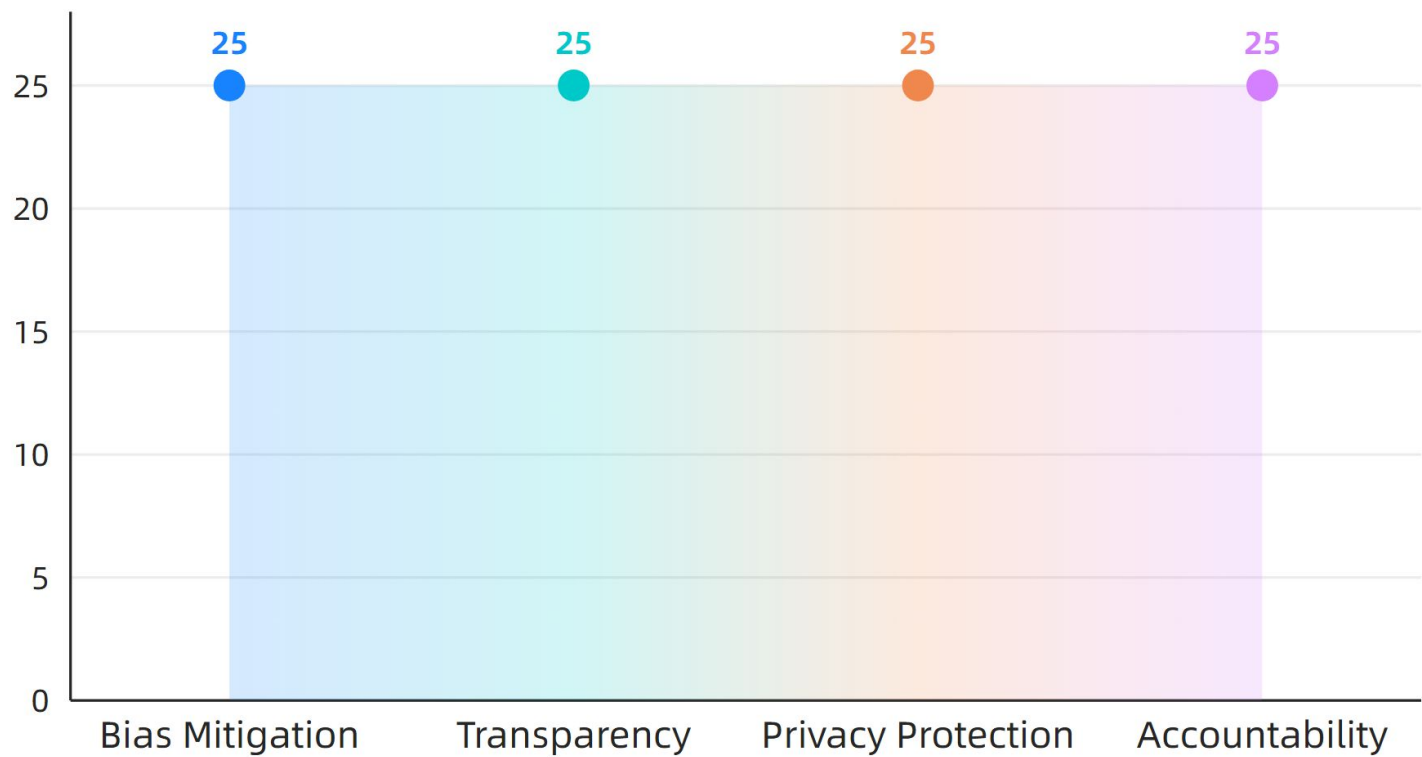
Implementation Process



Traditional vs Responsible AI

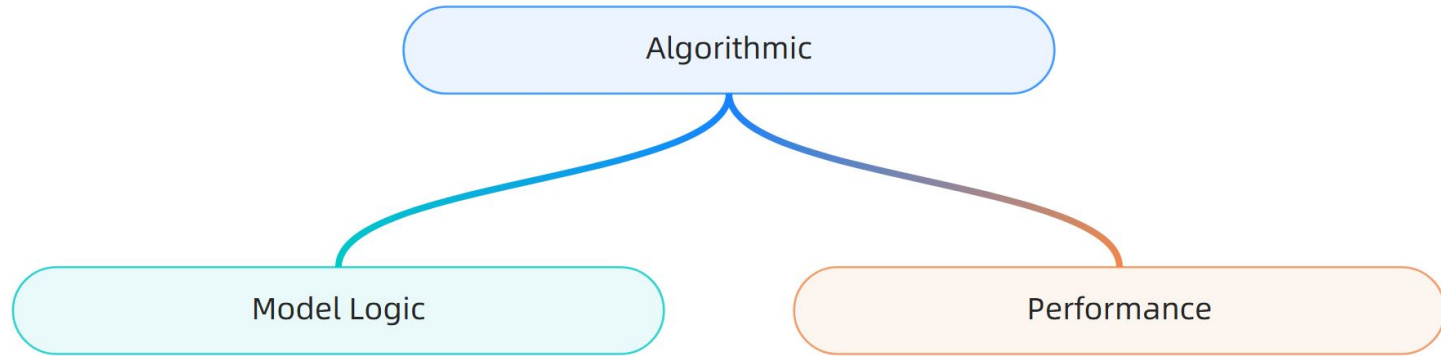


Core Responsibility Areas



Equal priority areas for responsible AI (Industry Analysis, 2024)

Transparency Strategies



Key Implementation Takeaways

Multi-Stakeholder

Collaborate across disciplines and roles

Continuous Monitoring

Regular audits and performance tracking

User Education

Stakeholder training and awareness programs

Global Standards

Follow OECD principles and frameworks

Activity: Champion best practices for responsible and ethical use of generative AI, addressing concerns about transparency, bias, and societal impact

- Activity: Ethical AI Champion Action Plan
- Scenario: Your organization is implementing a new generative AI chatbot for customer service. As the AI ethics champion, develop a comprehensive responsible deployment strategy addressing transparency, bias, and societal impact concerns.
- Step 1: Assess current AI governance structure and identify gaps
- Step 2: Design transparency framework covering algorithmic, data, process, and interpretability elements
- Step 3: Create bias mitigation plan including diverse data sources and fairness techniques
- Step 4: Establish monitoring and audit schedule with multi-disciplinary team
- Step 5: Develop stakeholder education program and public communication strategy
- Expected Output: Complete responsible AI implementation roadmap with timelines, accountability measures, and success metrics

Summary & Q&A

TRAQOM Survey (Mandatory)

It is mandatory to fill up the TRAQOM survey for WSQ funded course. You can find the QR code and course run ID from the LMS <https://ai-lms-tms.tertiaryinfo.tech/>



Certificate Delivery (Mandatory)

Please provide the details of your name and email to facilitate the issuance of your certificate after the class. You can find the QR code and Course Ref Cod from the LMS

<https://ai-lms-tms.tertiaryinfo.tech/>



CERTIFICATE OF ACCOMPLISHMENT

This Certificate is proudly presented to

[Student Name]

For completing the course

[Course Name]

Held on **[Course Dates]**

Dr. Alfred Ang

Managing Director
Tertiary Infotech Academy



Digital Attendance

- It is mandatory for you to take both AM, PM and Assessment digital attendance.
- Please scan the QR code from your mobile phone camera and submit your attendance.
- Ensure your attendance is recorded for all course days.
- This is required for funding and certification purposes.

Final Assessment

Support

- If you have any enquiries during and after the class, you can contact us below
 - Email: enquiry@tertiaryinfotech.com
 - Tel: +65 6318 4588
 - Website: www.tertiarycourses.com.sg

Thank You!